



Automorphisms of the Pants Graph of a Nonorientable Surface

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Abstract

We prove that, except in certain low-complexity cases, the automorphism group of the graph of pants decompositions of a nonorientable surface is isomorphic to the mapping class group of that surface.

Keywords Mapping class group · Nonorientable surface · Pants decomposition · Curve complex

1 Introduction

Let S be a compact surface, possibly nonorientable and with boundary. We define the mapping class group $\text{Mod}(S)$ of S to be the group of isotopy classes of all homeomorphisms of S . If S is orientable then $\text{Mod}(S)$ is usually called the extended mapping class group and we denote by $\text{Mod}^+(S)$ the subgroup of $\text{Mod}(S)$ consisting of orientation preserving mapping classes.

A connected orientable (resp. nonorientable) surface of genus g with b boundary components will be denoted by $S_{g,b}$ (resp. $N_{g,b}$). If $b = 0$ we drop it from the notation. Note that $N_{g,b}$ is obtained from $S_{0,g+b}$ by gluing g Möbius bands (also called crosscaps) along g distinct boundary components of $S_{0,g+b}$.

One of the most important breakthroughs in the study of mapping class groups of orientable surfaces was the introduction of the so-called *cut-system complex* by Hatcher and Thurston [7]. For a closed orientable surface S_g , they proved that this 2-dimensional complex is connected and simply-connected. Moreover, the action of $\text{Mod}^+(S_g)$ on the cut-system complex is cocompact, so a finite presentation of $\text{Mod}^+(S_g)$ can be derived from presentations of stabilizers of vertices and edges. This

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program was initiated in [7], completed by Harer [4], and later substantially simplified by Wajnryb [16, 17]; see [17] for a detailed account.

1.1 The Pants Complex

In the appendix to [7], Hatcher and Thurston introduced another important concept – the graph $\mathcal{P}^1(S_{g,b})$ of maximal cut systems of an oriented surface. The vertices of this graph are pants decompositions of $S_{g,b}$ and edges correspond to two types of elementary moves (see Section 3). Later, Hatcher, Lochak and Schneps [5, 6] proved that $\mathcal{P}^1(S_{g,b})$ is connected. They also showed that, after attaching five families of 2-cells, one obtains a simply connected 2-dimensional complex $\mathcal{P}(S_{g,b})$, now called the *pants complex* of $S_{g,b}$.

There are a number of interesting results concerning $\mathcal{P}(S_{g,b})$. Brock [3] proved that $\mathcal{P}^1(S_{g,b})$ is quasi-isometric to Teichmüller space with the Weil-Petersson metric, while Margalit [12] proved the following characterization of $\text{Aut}(\mathcal{P}(S_{g,b}))$.

Theorem 1.1 (Margalit [12]) *The natural map $\theta: \text{Mod}(S_{g,b}) \rightarrow \text{Aut}(\mathcal{P}(S_{g,b}))$ is surjective. Moreover, $\ker \theta \cong \mathbb{Z}_2$ for $S \in \{S_{1,1}, S_{1,2}, S_2\}$, $\ker \theta \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2$ for $S = S_{0,4}$, $\ker \theta = \text{Mod}(S)$ for $S = S_{0,3}$, and $\ker \theta$ is trivial otherwise.*

Later, this result was extended by Aramayona [1] to injective simplicial and locally-injective simplicial maps of $\mathcal{P}(S)$.

The proof of Theorem 1.1 is based on a similar result for another simplicial complex associated to a surface, namely the *curve complex* $\text{Curv}(S)$. The vertices of $\text{Curv}(S)$ correspond to homotopy classes of nontrivial simple closed curves in S , and k -simplexes correspond to sets of $k + 1$ vertices that may be represented as a multicurve in S – see Section 2 for a proper definition. One of the fundamental properties of $\text{Curv}(S)$ is the following

Theorem 1.2 (Ivanov [8], Korkmaz [9], Luo [10]) *If $S \neq S_{0,3}$ is an orientable surface with $\chi(S) < 0$ and $\eta: \text{Mod}(S) \rightarrow \text{Aut}(\text{Curv}(S))$ is the natural map, then*

- (1) *η is surjective when $S \neq S_{1,2}$ and $\text{im } \eta$ is a proper subgroup of $\text{Aut}(\text{Curv}(S))$ that preserves the set of nonseparating curves for $S = S_{1,2}$.*
- (2) *$\ker \eta \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2$ for $S = S_{0,4}$, $\ker \eta \cong \mathbb{Z}_2$ for $S \in \{S_{1,1}, S_{1,2}, S_2\}$, and $\ker \eta$ is trivial otherwise.*

In contrast to the orientable case, much less is known about the pants complex of a nonorientable surface. The most significant result to date is due to Papadopoulos and Penner [13], who introduced two new elementary moves between pants decompositions of a nonorientable surface (see Section 3) and proved that the resulting pants graph $\mathcal{P}(N_{g,b})$ is connected.

1.2 The Main Theorems

The main goal of this paper is to extend Theorem 1.1 to the case of a nonorientable surface (Theorem 1.4). However, as a first step we need the following nonorientable version of Theorem 1.2.

Theorem 1.3 *If N is a nonorientable surface with $\chi(N) < 0$ and $\eta: \text{Mod}(N) \rightarrow \text{Aut}(\text{Curv}(N))$ is the natural map, then*

- (1) η is surjective;
- (2) $\ker \eta \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2$ for $N = N_{1,2}$, $\ker \eta \cong \mathbb{Z}_2$ for $N \in \{N_{2,1}, N_3\}$, and $\ker \eta$ is trivial otherwise.

The above theorem was proved by Atalan and Korkmaz [2] for nonorientable surfaces $N_{g,b}$ with $g + b \neq 4$, the case $N = N_{1,3}$ was settled in [15], and the proof for the remaining surfaces $N \in \{N_{2,2}, N_{3,1}, N_{4,0}\}$ is given in the present article (Theorem 9.1).

The following theorem is the main result of this paper.

Theorem 1.4 *If N is a nonorientable surface with $\chi(N) < 0$ and $\theta: \text{Mod}(N) \rightarrow \text{Aut}(\mathcal{P}(N))$ is the natural map, then*

- (1) θ is surjective;
- (2) $\ker \theta = \ker \eta$, where $\eta: \text{Mod}(N) \rightarrow \text{Aut}(\text{Curv}(N))$ is the map from Theorem 1.3.

As an immediate corollary, we get

Corollary 1.5 *If $N = N_{g,b}$ is a nonorientable surface with $\chi(N) < 0$, then*

- (1) $\text{Aut}(\mathcal{P}(N)) \cong \text{Aut}(\text{Curv}(N))$,
- (2) $\text{Aut}(\mathcal{P}(N)) \cong \text{Aut}(\text{Curv}(N)) \cong \text{Mod}(N)$, for $g + b \geq 4$.

We also prove that if two nonorientable surfaces have isomorphic pants graphs, then they are homeomorphic (see Theorem 8.4). Finally, we show that it is enough to add 3 types of 2-dimensional cells (two types of triangles and one type of pentagonal cells) to $\mathcal{P}(N_3)$ to obtain a connected and simply-connected complex $\overline{\mathcal{P}(N_3)} \supset \mathcal{P}(N_3)$ – see Theorem 10.15.

1.3 Comparison with the Case of an Orientable Surface

Our proof of Theorem 1.4 has a similar structure as the proof of Theorem 1.1 in [12]; however, there are some important differences.

If S is an oriented surface, then to each edge of $\mathcal{P}(S)$ there is an associated Farey graph, and these graphs are the main tool to control the action of elements of $\text{Aut}(\mathcal{P}(S))$. If N is nonorientable, then the situation is more complicated, because there are different kinds of subgraphs associated to edges in $\mathcal{P}(N)$ (see Proposition 4.23). Moreover, there are edges in $\mathcal{P}(N)$ that are not contained in any triangle (see Example 4.12).

Another new phenomenon in the nonorientable case is the fact that two pants decompositions of the same surface can have a different number of components (see for example Fig. 1). The existence of moves of type 4 (Fig. 2) implies that if N has a pants decomposition with $p \geq 2$ one-sided and q two-sided curves, then N has also a pants decomposition with $p - 2$ one-sided and $q + 1$ two-sided curves. This significantly complicates the notion of 2-tight paths in $\mathcal{P}(N)$ (corresponding to the notion of 2-curve small paths in [12]) – see Definition 5.5.

Finally, in an oriented case, the pants graph $\mathcal{P}^1(S)$ is embedded in a simply-connected pants complex $\mathcal{P}(S)$, and we have a short list of types of 2-cells in that complex. It turns out that these 2-dimensional cells play a crucial role in the proof of Theorem 1.1. On the other hand, if N is nonorientable, then we only have a connected graph $\mathcal{P}(N) = \mathcal{P}^1(N)$, and the full list of types of 2-dimensional cells has yet to be discovered. In fact, in the present paper we discovered three interesting 2-cells: a pentagon in $\mathcal{P}(N_3)$, a tame alternating hexagon in $\mathcal{P}(N_{1,3})$, and a standard heptagon in $\mathcal{P}(N_{2,2})$ (see Figs. 19, 11 and 8). We expect that these types of cells will be crucial in constructing the 2-dimensional structure of $\mathcal{P}(N)$, but we leave this for future work.

1.4 Outline of the Paper

In Section 3, we review the definition of the pants graph $\mathcal{P}(N)$ of a nonorientable surface $N = N_{g,b}$. We also identify this graph for surfaces with some small values of g and b . In Section 4, we identify the subgraphs of $\mathcal{P}(N)$ corresponding to edges of $\mathcal{P}(N)$ (Remark 4.7). We then classify the triangles in $\mathcal{P}(N)$ (Proposition 4.9) and we prove the abstract characterization of moves (Proposition 4.23). In Section 5, we study properties of short loops (circuits) in $\mathcal{P}(N)$ and we introduce the circuit corresponding to a 2-cell in $\mathcal{P}(N_{2,2})$ (the standard heptagon). The analysis of short circuits is continued in Section 6, where we prove that, in most cases, every edge of $\mathcal{P}(N)$ can be extended to a short circuit with some special properties (Proposition 6.4). This construction is a crucial step toward proving that there is a well-defined correspondence between $\text{Aut}(\mathcal{P}(N))$ and $\text{Aut}(\text{Curv}(N))$, which we establish in Section 7, Theorem 7.5. In the same section, we complete the proof of our main theorem, Theorem 1.4.

In Section 8, we prove that if two nonorientable surfaces have isomorphic pants graphs, then they are homeomorphic (Theorem 8.4). Section 9 contains the proof of Theorem 1.3 in the case $g + b = 4$ (Theorem 9.1). Finally, in Section 10, we focus on the case $N = N_3$. We describe the structure of $\mathcal{P}(N)$ and construct an abstract graph isomorphic to it. Additionally, we define a simply-connected 2-dimensional complex $\overline{\mathcal{P}(N)}$ whose 1-skeleton is $\mathcal{P}(N)$ and such that $\text{Aut}(\overline{\mathcal{P}(N)}) = \text{Aut}(\mathcal{P}(N))$ (Theorem 10.15).

2 Preliminaries

Let S be a connected compact surface of negative Euler characteristic. Recall that $\chi(S_{g,b}) = 2 - 2g - b$, whereas $\chi(N_{g,b}) = 2 - g - b$. A *non-trivial curve* on S is an embedded simple closed curve which does not bound a disc nor a Möbius band in S and which is not homotopic to a boundary component of S . By a curve we always understand a non-trivial curve. A curve is either two- or one-sided depending on whether its regular neighbourhood is an annulus or a Möbius band respectively. To simplify the notation, we identify a curve with its isotopy class. We denote by $i(\alpha, \beta)$ the geometric intersection number of two curves α, β . If $i(\alpha, \beta) = 0$ and $\alpha \neq \beta$, then we say that α and β are *disjoint*.

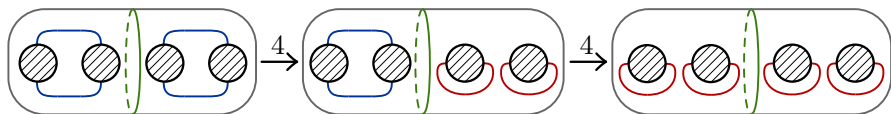


Fig. 1 Pants decompositions of N_4 with different cardinalities

A (finite) set of pairwise disjoint curves on S is called a *multicurve*. For a multicurve X we denote by X^+ (resp. X^-) the set of two-sided (resp. one-sided) curves of X , and by $|X|$ the number of curves in X .

The *curve complex* $\text{Curv}(S)$ is a simplicial complex whose k -simplices are multicurves of cardinality $k + 1$ on S . The k -skeleton of $\text{Curv}(S)$ is denoted by $\text{Curv}^k(S)$.

If $S = S_{0,4}$ or $S = S_{1,1}$, the definition of $\text{Curv}(S)$ is modified by declaring $\alpha, \beta \in \text{Curv}^0(S)$ to be adjacent in $\text{Curv}(S)$ whenever $i(\alpha, \beta) = 2$ or $i(\alpha, \beta) = 1$ respectively. The resulting connected graph is isomorphic to the *Farey graph* – see Section 3 of [11]. For the description of $\text{Curv}(N)$ for nonorientable surfaces of small complexity see [14].

For a multicurve X on S we denote by S_X the surface obtained by cutting S along X . A component of S_X is *trivial* if it is homeomorphic to $S_{0,3}$. A *pants decomposition* of S is a multicurve X such that all components of S_X are trivial. Note that a pants decomposition is a maximal multicurve.

Any pants decomposition of an orientable surface $S_{g,b}$ consists of $3g + b - 3$ curves. The number of curves in a pants decomposition of a nonorientable surface $N_{g,b}$ varies. For any integer m such that $0 \leq m \leq g$ and $m \equiv g \pmod{2}$ there is a pants decomposition of $N_{g,b}$ of cardinality

$$\frac{3g + m}{2} + b - 3$$

containing exactly m one-sided curves – see Fig. 1 (the shaded disks represent cross-caps, hence their interiors are to be removed and then the antipodal points on each boundary component are to be identified).

3 The Pants Graph

3.1 Definition and Connectivity

Let N be a nonorientable surface with $\chi(N) < 0$. We denote by $\mathcal{P}^0(N)$ the set of all pants decompositions of N . The *pants graph* of N , denoted by $\mathcal{P}(N)$ is a graph whose vertex set is $\mathcal{P}^0(N)$ and whose edges correspond to elementary moves between pants decompositions. An *elementary move* is an operation of replacing one curve α of a pants decomposition X either by a curve β , or by a pair of disjoint one-sided curves $\{\beta_1, \beta_2\}$ to obtain a different pants decomposition Y . There are four types of elementary move depending on the nontrivial component of $N_{X \setminus \{\alpha\}}$ called the *support* of the move. The intersection numbers of the involved curves are given in the following table (see also Fig. 2).

Type	Support	Notation	Curves	Intersection
1	$S_{1,1}$	$\alpha \xrightarrow{1} \beta$	$\alpha \in X^+, \beta \in Y^+$	$i(\alpha, \beta) = 1$
2	$S_{0,4}$	$\alpha \xrightarrow{2} \beta$	$\alpha \in X^+, \beta \in Y^+$	$i(\alpha, \beta) = 2$
3	$N_{1,2}$	$\alpha \xrightarrow{3} \beta$	$\alpha \in X^-, \beta \in Y^-$	$i(\alpha, \beta) = 1$
4	$N_{2,1}$	$\alpha \xrightarrow{4} \{\beta_1, \beta_2\}$	$\alpha \in X^+, \{\beta_1, \beta_2\} \subseteq Y^-$	$i(\alpha, \beta_1) = i(\alpha, \beta_2) = 1$

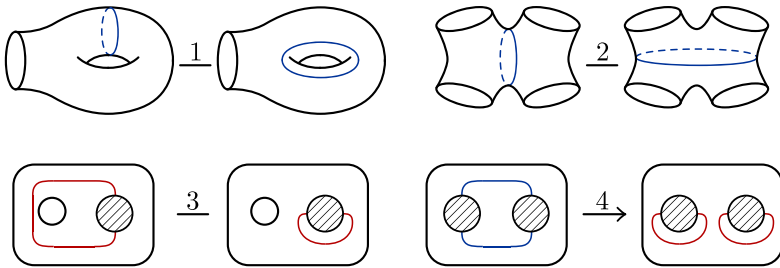


Fig. 2 The four elementary moves between pants decompositions

A move of type 4 has a natural direction towards the pants decomposition of a bigger cardinality, which we indicate by an arrow. For the sake of symmetry, the inverse of such a move, replacing a pair of disjoint one-sided curves by a single two-sided curve, will also be considered a move of type 4.

The pants graph of an orientable surface is defined analogously, using only moves of type 1 and 2 coming from the seminal paper [7]. Moves of type 3 and 4, supported on nonorientable surfaces and involving one-sided curves, were introduced by Papadopoulos and Penner [13] who proved the following fundamental result.

Theorem 3.1 ([13]) *For every nonorientable surface N with $\chi(N) < 0$ the pants graph $\mathcal{P}(N)$ is connected.*

3.2 Notation for Edges

We denote by XY an edge of $\mathcal{P}(N)$ (elementary move) joining vertices X and Y . We also write

$$X \xrightarrow{i} Y \quad \text{or} \quad XY: \alpha \xrightarrow{i} \beta \quad \text{or} \quad \alpha \xrightarrow{i} \beta$$

to denote a move of type i replacing $\alpha \in X$ by $\beta \in Y$. The type of a move may be dropped if it is arbitrary or unknown. A move of type 4, changing the number of curves in a pants decomposition, has a natural direction towards the vertex of a bigger cardinality, which we indicate by an arrow

$$X \xrightarrow{4} Y \quad \text{or} \quad XY: \alpha \xrightarrow{4} \{\beta_1, \beta_2\} \quad \text{or} \quad \alpha \xrightarrow{4} \{\beta_1, \beta_2\}.$$

Unless explicitly stated otherwise, letters of the Greek alphabet denote curves. An underlined Greek letter, e.g. $\underline{\alpha}$, denotes either a curve, or a multicurve consisting of

two one-sided curves. For example

$$XY: \alpha \text{ --- } \underline{\alpha'}$$

denotes a move replacing a curve α by $\underline{\alpha'}$, which is either a single curve, or a pair of one-sided curves. In the latter case XY is a move of type 4 directed from X to Y .

3.3 Pants Graph of $S_{0,4}$, $S_{1,1}$, $N_{1,2}$ and $N_{2,1}$

The graphs $\mathcal{P}(S_{0,4})$ and $\mathcal{P}(S_{1,1})$ are the same as the curve complexes of these surfaces and are isomorphic to the Farey graph – see Section 3 of [11].

Proposition 3.2 (*Pants graph of $N_{1,2}$, [13]*) *The graph $\mathcal{P}(N_{1,2})$ consists of two vertices, corresponding to two one-sided curves, joined by an edge of type 3.*

Definition 3.3 (Infinite fan) *The infinite fan F_∞ is the graph with vertex set $\{v\} \cup \{v_i\}_{i \in \mathbb{Z}}$ and edge set $\{vv_i, v_i v_{i+1}\}_{i \in \mathbb{Z}}$. The vertex v is called the *centre* of F_∞ .*

Proposition 3.4 (*Pants graph of $N_{2,1}$, [13]*) *The graph $\mathcal{P}(N_{2,1})$ is isomorphic to F_∞ . Every isomorphism $F_\infty \rightarrow \mathcal{P}(N_{2,1})$ maps the centre of F_∞ to the pants decomposition consisting of the unique non-separating two-sided curve in $N_{2,1}$.*

Proof We only show how to define an isomorphism $F_\infty \rightarrow \mathcal{P}(N_{2,1})$, see [13] for details. Let α be the unique non-separating two-sided curve in $N = N_{2,1}$ and let $T_\alpha \in \text{Mod}(N)$ be a Dehn twist about α . We have $\{\alpha\} \in \mathcal{P}^0(N)$. Choose a one-sided curve β in N . Since $N_\beta \cong N_{1,2}$, by Proposition 3.2 there are exactly two pants decompositions of N containing β , namely $\{\beta, T_\alpha(\beta)\}$ and $\{\beta, T_\alpha^{-1}(\beta)\}$, joined in $\mathcal{P}(N)$ by an edge of type 3. Each of these vertices is also connected to $\{\alpha\}$ by an edge of type 4. The map $\phi: F_\infty \rightarrow \mathcal{P}(N)$ defined by

$$\phi(v) = \{\alpha\}, \quad \phi(v_i) = \{T_\alpha^i(\beta), T_\alpha^{i+1}(\beta)\}, i \in \mathbb{Z}$$

is an isomorphisms of graphs. $\square$

4 Farey and Fan Subgraphs

Definition 4.1 For $X \in \mathcal{P}(N)$ and $\alpha \in X$ we denote by $F(X, \alpha)$ the subgraph of $\mathcal{P}(N)$ spanned by all the pants decompositions containing $X \setminus \{\alpha\}$.

Remark 4.2 For $X \in \mathcal{P}(N)$ and $\alpha \in X$ let M be the non-trivial (i.e. $M \not\cong S_{0,3}$) component of $N_{X \setminus \{\alpha\}}$. Then $F(X, \alpha) \cong \mathcal{P}(M)$ and if $\alpha \in X^+$, then $M \in \{S_{0,4}, S_{1,1}, N_{2,1}\}$, whereas if $\alpha \in X^-$, then $M \cong N_{1,2}$. Therefore, we have the following characterization of the subgraphs $F(X, \alpha)$.

(1) If $\alpha \in X^+$ and $M \cong S_{0,4}$ or $M \cong S_{1,1}$, then $F(X, \alpha)$ is isomorphic to the Farey graph.

- (2) If $\alpha \in X^+$ and $M \cong N_{2,1}$, then $F(X, \alpha)$ is isomorphic to the infinite fan F_∞ by Proposition 3.4. Note that in this case X is the unique vertex of infinite degree in $F(X, \alpha)$ and we call it the *centre* of $F(X, \alpha)$.
- (3) If $\alpha \in X^-$, then $F(X, \alpha)$ consist of two vertices joined by an edge of type 3 replacing α by a uniquely determined α' by Proposition 3.2.

Definition 4.3 Let $X \in \mathcal{P}(N)$ and $\alpha \in X^+$. We say that α is *special* in X if the non-trivial component of $N_{X \setminus \{\alpha\}}$ is a one-holed Klein bottle $N_{2,1}$.

Remark 4.4 By Remark 4.2, $F(X, \alpha) \cong F_\infty$ if and only if α is special in X . Moreover, if α is special in X , then X is the centre of $F(X, \alpha)$.

Remark 4.5 By the previous remark, if

$$XY: \alpha \xrightarrow{4} \{\beta, \beta'\}$$

is a move of type 4, then α is special in X , $\alpha \notin Y$ and XY is an edge of $F(X, \alpha) \cong F_\infty$. Moreover, X is of infinite degree in $F(X, \alpha)$.

Definition 4.6 Let XY be an edge of $\mathcal{P}(N)$. We denote by $F(XY)$ the subgraph of $\mathcal{P}(N)$ spanned by all pants decompositions containing $X \cap Y$.

Remark 4.7 If $XY: \alpha \xrightarrow{\quad} \alpha'$ then $F(XY) = F(X, \alpha)$. By Remarks 4.2–4.5, $F(XY)$ is isomorphic

- (1) to the Farey graph if XY is of type 1 or 2;
- (2) to the infinite fan F_∞ if XY is of type 4;
- (3) to the single edge graph if XY is of type 3.

Lemma 4.8 If XY is a move of type 1, 2 or 3 and γ is special in X , then $\gamma \in Y$.

Proof Suppose $\gamma \notin Y$. Then XY must replace γ by some $\delta \in Y$. By definition $\gamma \in X^+$, hence $\delta \in Y^+$ and $i(\gamma, \delta) > 0$. Let K be the one-holed Klein bottle component of $N_{X \setminus \{\gamma\}} = N_{Y \setminus \{\delta\}}$. Since γ is the unique two-sided curve in K we have $\gamma = \delta$ which is a contradiction. $\square$

Proposition 4.9 (Classification of triangles) Every triangle in $\mathcal{P}(N)$ is of one of the two forms. Either

$$(1) \quad \begin{array}{c} \alpha' \\ \swarrow \quad \searrow \\ \alpha \quad \quad \quad \alpha'' \\ \nwarrow \quad \nearrow \\ \alpha \end{array}$$

where $i \in \{1, 2\}$, or

$$(2) \quad \begin{array}{c} \{\beta_1, \beta_2\} \\ \nearrow \quad \searrow \\ \alpha \quad \quad \quad \alpha \\ \nwarrow \quad \nearrow \\ \{\beta'_1, \beta'_2\} \end{array}$$

A triangle of the form (1) (resp. (2)) will be called a Farey triangle (resp. fan triangle).

Proof Let $X, Y, Z \in \mathcal{P}(N)$ be vertices of a triangle.

Suppose first that $|X| = |Y| = |Z| = n$ and XY is a move replacing $\alpha \in X$ by $\alpha' \in Y$. We have $i(\alpha, \alpha') > 0$ so $\alpha \notin Z$ or $\alpha' \notin Z$. But $|X \cap Z| = |Y \cap Z| = n - 1$ which implies $X \cap Z = Y \cap Z = X \setminus \{\alpha\}$. We see that XYZ is of the form (1) but we still need to show that $i \neq 3$. Suppose that α is one-sided. The non-trivial component of $N_{X \setminus \{\alpha\}}$ is homeomorphic to $N_{1,2}$. This component contains α, α' and α'' , which is a contradiction, because there are only two isotopy classes of nontrivial curves in $N_{1,2}$ – see Proposition 3.2.

Now suppose $n = |X| < |Y|$ and XY is a move of type 4 replacing $\alpha \in X^+$ by $\beta_1, \beta_2 \in Y^-$. We have $n \leq |Z| \leq n + 1$ so we consider two cases. First suppose $|Z| = n$ so that XZ is a move of type 1, 2 or 3, and ZY is a move of type 4. Since α is special in X , by Lemma 4.8, $\alpha \in Z$. We see that ZY replaces α by β_1, β_2 so $ZY = XY$ and $X = Z$, a contradiction. Therefore, $|Z| = n + 1$ and XZ is a move of type 4. Now $|Z \cap Y| = n$ and we can assume that $\beta_2 \in Z$. We have $\alpha \notin Z$ because $i(\beta_2, \alpha) = 1$ so XZ replaces α by β'_1, β_2 for some one-sided $\beta'_1, \beta'_1 \neq \beta_1$. It follows that YZ is a move of type 3 replacing β_1 by β'_1 . $\square$

Corollary 4.10 *Let XYZ be a triangle in $\mathcal{P}(N)$. If XYZ is a fan triangle, assume that YZ is of type 3. Then:*

- (1) *there exists a unique $\alpha \in X^+$ such that*

$$XYZ \subset F(X, \alpha) = F(XY);$$

- (2) *if XYZ' is another triangle in $\mathcal{P}(N)$ satisfying the same assumption (namely, if XYZ' is a fan triangle, then and YZ' is of type 3), then*

$$XYZ' \subset F(X, \alpha) = F(XY),$$

where α is the curve from (1).

Proof If $XY: \alpha \xrightarrow{\quad} \alpha'$, then Proposition 4.9 implies that

$$XYZ \subset F(X, \alpha)$$

$$XYZ' \subset F(X, \alpha).$$

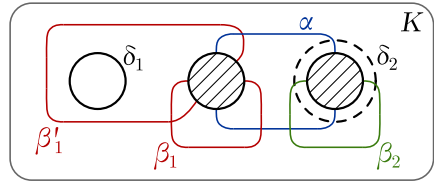
Combine this with Definition 4.6 to get the statement. $\square$

Lemma 4.11 *Suppose that XY is an edge of type 3 in $\mathcal{P}(N)$.*

- (1) *If $N \neq N_3$, then XY belongs to at most one triangle.*
 (2) *If $N = N_3$, then XY either does not belong to any triangle, or it belongs to exactly two different triangles. It belongs to two triangles if and only if X and Y are vertices of degree 6 in $\mathcal{P}(N)$.*

Proof Let $XY: \beta_1 \xrightarrow{3} \beta'_1$. By Lemma 4.9, an edge of type 3 can belong only to a fan triangle, and XY belongs to such a triangle if and only if there is a multicurve $\gamma \subset (X \setminus \beta_1) = (Y \setminus \beta'_1)$ such that the only non-trivial component of N_γ is a one-holed

Fig. 3 Curves of the fan triangle in $N_{2,1}$



Klein bottle K containing β_1 and β'_1 . Such γ uniquely determines a fan triangle XYZ of the form

$$ZX: \alpha \xrightarrow{4} \{\beta_1, \beta_2\}, \quad XY: \beta_1 \xrightarrow{3} \beta'_1, \quad YZ: \{\beta'_1, \beta_2\} \xleftarrow{4} \alpha.$$

Indeed, α is the unique two-sided curve in K and $\beta_2 \in X^- \cap Y^-$ is the unique one-sided curve in K disjoint from $\beta_1 \cup \beta'_1$ – see Fig. 3.

Suppose that such γ exists. The regular neighbourhood of $\beta_1 \cup \beta'_1$ is a projective plane $N_{1,2}$ with two boundary components δ_1, δ_2 and the existence of γ implies that at least one of these two boundary components, say δ_2 , cuts a projective plane from N (Fig. 3). Therefore, if $N \neq N_3$, then γ is unique: $\gamma = X \setminus \{\beta_1, \beta_2\}$.

If $N = N_3$, then γ is a single one-sided curve disjoint from $\beta_1 \cup \beta'_1$, and there are two such curves (Fig. 4).

In this case each of the vertices X and Y is a pants decomposition consisting of three one-sided curves. Such a vertex has degree 6, as it is incident to three edges of type 3 and three edges of type 4 (Fig. 5). $\square$

Example 4.12 Let

$$XY: \{\alpha, \beta\} \xrightarrow{3} \{\alpha, \beta'\}$$

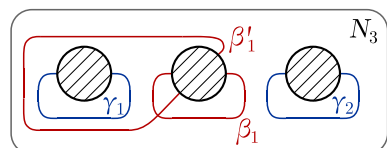
be an edge of type 3 in $\mathcal{P}(N_3)$, where α, β, β' are as in Fig. 6.

Since the complement of β in N_3 is orientable, this edge can not be contained in a fan triangle. Moreover, this edge cannot be contained in a Farey triangle, because β is one-sided. By Proposition 4.9, XY is not contained in any triangle.

Definition 4.13 Let F be a subgraph of $\mathcal{P}(N)$. We say that

- (1) F is a *Farey subgraph* if F is isomorphic to the Farey graph;
- (2) F is a *fan subgraph* if F is isomorphic to the infinite fan F_∞ and F is not contained in any Farey subgraph;
- (3) F is a *thin edge* if F consists of a single edge, F is not contained in any Farey subgraph and if F is contained in a fan subgraph F' , then both vertices of F are of finite degree in F' .

Fig. 4 One-sided curves in N_3



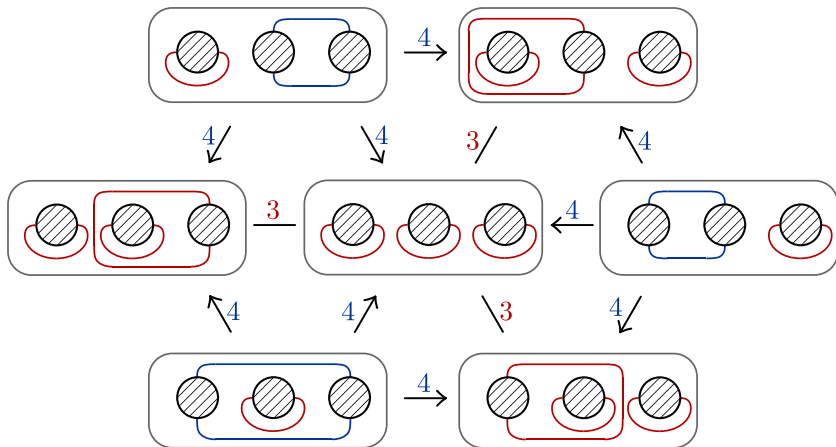


Fig. 5 The vertex of degree 6 in $\mathcal{P}(N_3)$

Remark 4.14 Note that every Farey subgraph contains subgraphs that are isomorphic to F_∞ . However, by the above definition, they are not fan subgraphs of $\mathcal{P}(N)$.

Lemma 4.15 Let F be an edge of type 3 in $\mathcal{P}(N)$. Then F is a thin edge.

Proof Every edge in a Farey subgraph F' belongs to two different triangles in F' and every vertex of F' has infinite degree in F' . Therefore, by Lemma 4.11, F can not be contained in F' .

If F is contained in a fan subgraph F' and one of the vertices of F is of infinite degree in F' , then F belongs to two different triangles in F' . This contradicts Lemma 4.11. $\square$

Lemma 4.16 If

$$XY: \alpha \xrightarrow{4} \{\beta_1, \beta_2\}$$

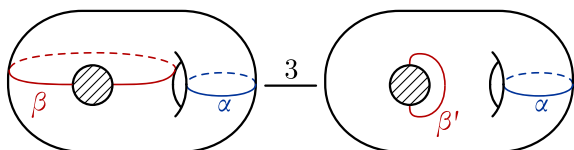
is an edge of type 4 in $\mathcal{P}(N)$, then the fan graph $F(XY) = F(X, \alpha)$ is not contained in any Farey subgraph, so it is a fan subgraph of $\mathcal{P}(N)$.

Proof Recall that $F(XY) = F(X, \alpha) \cong \mathcal{P}(N_{2,1})$ is a fan subgraph by Remarks 4.2 and 4.5. Moreover, by the proof of Proposition 3.4, $\mathcal{P}(N_{2,1})$ contains edges of type 3. By Lemma 4.15, $F(XY)$ can not be contained in any Farey subgraph. $\square$

Definition 4.17 A marked F -graph is a pair (F, X) , where X is a vertex of F and F is either

(1) a Farey subgraph of $\mathcal{P}(N)$, or

Fig. 6 An edge of type 3 in $\mathcal{P}(N_3)$ that is not contained in any triangle



- (2) a fan subgraph of $\mathcal{P}(N)$, and X is the centre of F , or else
 (3) a single thin edge in $\mathcal{P}(N)$.

Remark 4.18 If we combine Remark 4.7 and Lemmas 4.15 and 4.16, we see that to every oriented edge XY of $\mathcal{P}(N)$ with $|X| \leq |Y|$, there is an associated marked F -graph $(F(XY), X)$.

The following proposition shows that the above correspondence is surjective.

Proposition 4.19 *If (F, X) is a marked F -graph, then $F = F(X, \alpha)$ for a unique $\alpha \in X$. Moreover,*

- (1) $\alpha \in X^+$ if and only if F is a Farey or a fan subgraph;
 (2) α is special in X if and only if F is a fan subgraph.

Proof Suppose first that F is a Farey or fan subgraph of $\mathcal{P}(N)$ and let T be a triangle of F with vertices X, Y, Z , where X is of infinite degree in F (see Definition 4.17). By Proposition 4.9, T is either a Farey triangle or a fan triangle. In the latter case, since X is of infinite degree in F , each of the edges XY and XZ belongs to two different triangles in F , hence Lemma 4.11 implies that XY and XZ are moves of type 4. In either case T satisfies the assumptions of Corollary 4.10, so $T \subset F(X, \alpha)$ for a unique $\alpha \in X^+$.

Now, if T' is any triangle of F , then T' and T are connected by a sequence of triangles such that every two consecutive triangles in that sequence satisfy the assumptions of part (2) of Corollary 4.10. It follows that all these triangles are contained in $F(X, \alpha)$ and consequently $F \subseteq F(X, \alpha)$. Since every injective simplicial map $F \rightarrow F$ is an isomorphism, $F = F(X, \alpha)$.

Assume now that F is a thin edge:

$$XY: \alpha - \underline{\alpha'}$$

If XY is an edge of type 1 or 2, then F is contained in a Farey graph $F(XY) = F(X, \alpha)$. This contradicts the definition of a thin edge. Similarly, if XY is of type 4, then Remark 4.5 and Lemma 4.16 imply that X is of infinite degree in a fan subgraph $F(XY) = F(X, \alpha)$. This contradicts the definition of a thin edge. Hence, F is of type 3, $F = F(X, \alpha)$ (see Remark 4.2) and α is the unique element of X such that $i(\alpha, \alpha') > 0$.

Statements (1) and (2) follow from Remarks 4.2 and 4.4. $\square$

The construction of $F(X, \alpha)$ in the proof of the above proposition implies the following corollary.

Corollary 4.20 *If (F, X) is a marked F -graph and*

$$XY: \alpha - \underline{\alpha'}$$

is an edge of F , then $F = F(X, \alpha) = F(XY)$.

Corollary 4.21 *If F is a Farey (resp. fan) subgraph of $\mathcal{P}(N)$, then every triangle of F is a Farey (resp. fan) triangle.*

Proof Let $F = F(X, \alpha)$ be as in Proposition 4.19. By Remark 4.2, all triangles of $F(X, \alpha)$ are either fan or Farey depending on whether $\alpha \in X^+$ is special or not. $\square$

Corollary 4.22 *If N, N' are nonorientable surfaces of negative Euler characteristic and $A: \mathcal{P}(N) \rightarrow \mathcal{P}(N')$ is an isomorphism, then for any edge XY of $\mathcal{P}(N)$*

$$A(F(XY)) = F(A(X)A(Y)).$$

Proof Assume $|X| \leq |Y|$. By Remark 4.18, $(F(XY), X)$ is a marked F-graph, so

$$(A(F(XY)), A(X))$$

is also a marked F-graph (see Definitions 4.13 and 4.17). Moreover, $A(X)A(Y)$ is and edge of $A(F(XY))$, so Corollary 4.20 implies that

$$A(F(XY)) = F(A(X)A(Y)).$$

$\square$

Proposition 4.23 (Characterisation of moves) *Let XY be an edge of $\mathcal{P}(N)$.*

- (1) *XY is a move of type 1 or 2 if and only if $F(XY)$ is a Farey subgraph;*
- (2) *XY is a move of type 4 directed from X to Y if and only if $F(XY)$ is a fan subgraph with centre X ;*
- (3) *XY is a move of type 3 if and only if $F(XY)$ is a thin edge.*

Proof Assume $|X| \leq |Y|$. By Remark 4.18, $(F(XY), X)$ is a marked F-graph and its type is uniquely determined by the type of XY by Remark 4.7.

Conversely, if $F(XY)$ is a Farey subgraph, then Corollary 4.21 implies that XY is of type 1 or 2. If $F(XY)$ is a fan subgraph with centre X , then Corollary 4.21 implies that XY is of type 3 or 4. But we also know that X is of infinite degree in $F(XY)$, so Lemma 4.15 implies that XY is an edge of type 4 and $|X| < |Y|$. Finally, if $F(XY)$ is a thin edge, then XY can not be of types 1, 2 or 4 by (1) and (2). $\square$

Remark 4.24 Note that there are two different types of edges of type 3 – some of them are contained in some fan subgraph of $\mathcal{P}(N)$, but some of them are not – see Example 4.12.

Corollary 4.25 *If N, N' are nonorientable surfaces of negative Euler characteristic and $A: \mathcal{P}(N) \rightarrow \mathcal{P}(N')$ is an isomorphism, then A maps an edge of type 3 or 4 to an edge of the same type. Furthermore, A also preserves directions of edges of type 4.*

Proof By Corollary 4.22, if XY is an edge of type 3 or 4 in $\mathcal{P}(N)$, then

$$F(A(X)A(Y)) = A(F(XY)) \cong F(XY),$$

so $A(X)A(Y)$ and XY are of the same type by Proposition 4.23. Furthermore, if $|X| < |Y|$ then X is the centre of $F(XY)$, so $A(X)$ is the centre of $A(F(XY))$ and $|A(X)| < |A(Y)|$. $\square$

5 Small Circuits

Definition 5.1 A *circuit* (resp. a *path*) is a subgraph of $\mathcal{P}(N)$ homeomorphic to a circle (resp. an interval). The *length* of a circuit (resp. a path) is the number of its edges.

We use the names *triangle*, *quadrangle*, *pentagon*, *hexagon*, *heptagon* for circuits of length at most 7.

Definition 5.2 A path or a circuit is *alternating* if no two consecutive edges (i.e. edges incident to a common vertex) are contained in one Farey or fan subgraph.

Remark 5.3 Alternating paths (resp. circuits) are mapped to alternating paths (resp. circuits) by automorphisms of $\mathcal{P}(N)$.

Lemma 5.4 (1) *Every alternating path of length 3 is of the form*

$$\{\underline{\alpha}_1, \underline{\alpha}_2\} - \{\underline{\alpha}_1', \underline{\alpha}_2\} - \{\underline{\alpha}_1', \underline{\alpha}_2'\}. \quad (5.1)$$

(2) *If P is a path of the form (5.1), then either P is alternating, or P is contained in a fan subgraph and both edges of P are of type 3.*

Recall that according to our convention, each of $\underline{\alpha}_1, \underline{\alpha}_1', \underline{\alpha}_2, \underline{\alpha}_2'$ may be a pair of one-sided curves corresponding to a move of type 4.

Proof Let P be a path $X - Y - Z$.

- (1) Suppose that P is alternating, $XY: \underline{\alpha}_1 - \underline{\alpha}_1'$, and $\underline{\alpha}_1' \notin Z$. If $\underline{\alpha}_1' = \alpha_1'$ is a single curve, then the path is contained in $F(Y, \alpha_1')$, and since it has length 3, α_1' is two-sided and $F(Y, \alpha_1')$ is either a Farey subgraph or a fan subgraph by Remark 4.2. This is a contradiction, because we assumed that the path is alternating. If $\underline{\alpha}_1' = \{\beta_1, \beta_2\}$ is a multicurve, then YZ is an edge of type 3 replacing one of the curves β_1 or β_2 . It follows that the path is contained in the fan subgraph $F(X, \alpha_1)$, which is also a contradiction. Therefore $\underline{\alpha}_1' \in Z$, which implies that the path is of the form (5.1).
- (2) Suppose that P is of the form (5.1) and $P \subset F$, where F is either a Farey or a fan subgraph. If Y has infinite degree in F , then (F, Y) is a marked F -graph and by Lemma 4.19 we have $F = F(Y, \alpha)$ for a unique $\alpha \in Y$. Since $X, Z \in F$ we have $\alpha_1' = \alpha = \alpha_2$, which is a contradiction. Thus Y has finite degree in F , which means that F is a fan subgraph and both edges of P are of type 3. $\square$

Definition 5.5 A subgraph Γ (path or circuit) of $\mathcal{P}(N)$ is *2-tight* if there exists $X_0 \in \Gamma^0$ such that

$$\left| \bigcap_{X \in \Gamma^0} X \right| \geq |X_0| - 2.$$

Remark 5.6 Note that a subpath of a 2-tight path may not be 2-tight. For example, the following circuit of length 4

$$\begin{array}{ccc} \{\alpha, \beta\} & \xrightarrow{4} & \{\alpha, \beta_1, \beta_2\} \\ 4 \downarrow & & \downarrow 4 \\ \{\alpha_1, \alpha_2, \beta\} & \xrightarrow{4} & \{\alpha_1, \alpha_2, \beta_1, \beta_2\} \end{array}$$

is 2-tight, but its subpath

$$\{\alpha_1, \alpha_2, \beta\} \xrightarrow{4} \{\alpha_1, \alpha_2, \beta_1, \beta_2\} \xleftarrow{4} \{\alpha, \beta_1, \beta_2\}$$

is not 2-tight.

Definition 5.7 A vertex X of a subgraph Γ of $\mathcal{P}(N)$ is *minimal* in Γ if $|X|$ is minimal among all vertices of Γ .

Lemma 5.8 Automorphisms of $\mathcal{P}(N)$ map minimal vertices of a subgraph Γ to minimal vertices of the image of Γ .

Proof A vertex X is minimal in Γ if and only if for every path from X to any other vertex Y of Γ , the number of edges of type 4 directed from X to Y is greater than the number of edges of type 4 directed from Y to X . By Corollary 4.25 this property is preserved by automorphisms of $\mathcal{P}(N)$. $\square$

Lemma 5.9 Suppose that P is an alternating and 2-tight path

$$W — X — Y — Z,$$

such that X is minimal in P . Then P is of the form

$$\{\underline{\alpha}_1', \alpha_2\} — \{\alpha_1, \alpha_2\} — \{\alpha_1, \underline{\alpha}_2'\} — \{\underline{\alpha}_1'', \underline{\alpha}_2'\}. \quad (5.2)$$

Proof By Lemma 5.4, the path $W — X — Y$ is of the form

$$\{\underline{\alpha}_1', \alpha_2\} — \{\alpha_1, \alpha_2\} — \{\alpha_1, \underline{\alpha}_2'\},$$

where α_1, α_2 are curves (because X is minimal). Since P is 2-tight we have

$$X \setminus \{\alpha_1, \alpha_2\} = Y \setminus \{\alpha_1, \underline{\alpha}_2'\} \subset Z,$$

and by applying Lemma 5.4 to the path $X — Y — Z$ we obtain that it is of the form

$$\{\alpha_1, \alpha_2\} — \{\alpha_1, \underline{\alpha}_2'\} — \{\underline{\alpha}_1'', \underline{\alpha}_2'\}.$$

$\square$

Lemma 5.10 *Every quadrangle is 2-tight.*

Proof Let $\mathcal{C}$ be a quadrangle $WXYZ$ and assume that X is minimal in $\mathcal{C}$. We assume $|X| \geq 3$, for otherwise $\mathcal{C}$ is obviously 2-tight. Let $XY: \alpha_1 \text{---} \underline{\alpha_1}'$.

If $\alpha_1 \in W$, then the path $W \text{---} X \text{---} Y$ is of the form

$$\{\alpha_1, \underline{\alpha_2}', \alpha_3\} \text{---} \{\alpha_1, \alpha_2, \alpha_3\} \text{---} \{\underline{\alpha_1}', \alpha_2, \alpha_3\}.$$

Note that $\alpha_1 \notin Z$ and $\alpha_2 \notin Z$, for otherwise $Z = X$. If $\mathcal{C}$ is not 2-tight, then there is $\alpha_3 \in X \setminus \{\alpha_1, \alpha_2\}$ such that $\alpha_3 \notin Z$. Then ZW and YZ must be edges of type 4:

$$ZW: \beta \xleftarrow{4} \{\alpha_1, \alpha_3\}, \quad YZ: \{\alpha_2, \alpha_3\} \xrightarrow{4} \beta$$

for some $\beta \in Z$. But then $\underline{\alpha_1}' = \alpha_1'$ and $\underline{\alpha_2}' = \alpha_2'$ are curves and

$$Z = \{\alpha_2', \beta, \dots\} = \{\alpha_1', \beta, \dots\},$$

which gives $\alpha_1' = \alpha_2'$. This is a contradiction, as $i(\alpha_1', \alpha_2) = 0 < i(\alpha_2', \alpha_2)$.

Now suppose $\alpha_1 \notin W$, so the path

$$W \text{---} X \text{---} Y$$

is of the form:

$$\{\underline{\alpha_1}'', \alpha_2, \alpha_3\} \text{---} \{\alpha_1, \alpha_2, \alpha_3\} \text{---} \{\underline{\alpha_1}', \alpha_2, \alpha_3\}.$$

If $\mathcal{C}$ is not 2-tight, then there are $\alpha_2, \alpha_3 \in X \setminus \{\alpha_1\}$ such that $\alpha_2, \alpha_3 \notin Z$. Then ZW and YZ must be edges of type 4:

$$ZW: \beta \xleftarrow{4} \{\alpha_2, \alpha_3\}, \quad YZ: \{\alpha_2, \alpha_3\} \xrightarrow{4} \beta$$

for some $\beta \in Z$. But then

$$Z = \{\underline{\alpha_1}'', \beta, \dots\} = \{\underline{\alpha_1}', \beta, \dots\},$$

which gives $\underline{\alpha_1}' = \underline{\alpha_1}''$ and $W = Y$, a contradiction. $\square$

Example 5.11 If $g \geq 4$, then there are pentagons in $\mathcal{P}(N_g)$ that are not 2-tight. For example, consider the pentagon of the form

$$\begin{array}{ccc} & & \{\beta_1, \beta_2, \gamma_1, \gamma_2\} \xleftarrow{4} \{\beta_1, \beta_2, \gamma\} \\ & \nearrow 4 & \\ \{\beta, \gamma_1, \gamma_2\} & & \\ & \searrow 4 & \\ & & \{\beta_1, \beta_2', \gamma_1, \gamma_2\} \xleftarrow{4} \{\beta_1, \beta_2', \gamma\} \end{array} \quad \begin{array}{c} \\ \\ 3 \\ \end{array}$$

where $\beta_1, \beta_2, \beta, \gamma_1, \gamma_2, \gamma$ are as in Fig. 7.

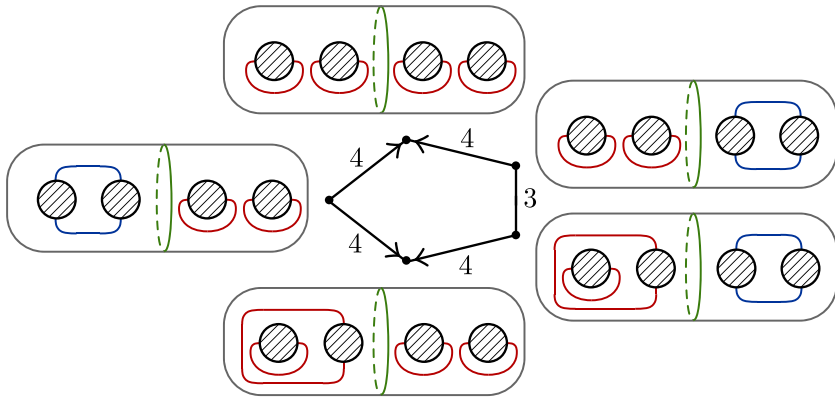


Fig. 7 Pentagon in $\mathcal{P}(N_4)$ that is not 2-tight

Lemma 5.12 *Let $\mathcal{C}$ be circuit of length at most 6 without edges of type 4. If $\mathcal{C}$ is not 2-tight, then it is an alternating hexagon with even number of edges of type 3.*

Proof If $\mathcal{C}$ does not contain edges of type 4, and we assume that $\mathcal{C}$ is not 2-tight, then the proof of Lemma 4 in [12] works verbatim and we conclude that $\mathcal{C}$ is an alternating hexagon with edges of the form

$$\begin{aligned} \alpha_1 &— \alpha'_1, \alpha_2 &— \alpha'_2, \alpha_3 &— \alpha'_3, \\ \alpha'_1 &— \alpha_1, \alpha'_2 &— \alpha_2, \alpha'_3 &— \alpha_3. \end{aligned}$$

Hence, the number of edges of type 3 is even – it is twice the number of one-sided curves among $\{\alpha_1, \alpha_2, \alpha_3\}$. $\square$

Definition 5.13 Let $N \notin \{N_{2,1}, N_3\}$ and suppose that $XY: \alpha \xrightarrow{4} \{\beta_1, \beta_2\}$ is a move of type 4. There is a unique $\gamma \in X \cap Y$ separating a one-holed Klein bottle containing α , β_1 and β_2 . We call it *the separating curve associated to XY* .

Remark 5.14 The separating curve associated to $XY: \alpha \xrightarrow{4} \{\beta_1, \beta_2\}$ is isotopic to the boundary of a regular neighbourhood of $\alpha \cup \beta_1$. In particular, it is determined by α and β_1 .

Lemma 5.15 *Suppose $N \notin \{N_{2,1}, N_3\}$ and P is a path in $\mathcal{P}(N)$ of the form*

$$X \xrightarrow{4} Y — Z,$$

where YZ is of type 1 or 2. Then Z contains the separating curve associated to XY if and only if the path P is contained in a quadrangle.

Proof Let

$$XY: \alpha_1 \xrightarrow{4} \{\beta_1, \beta_2\}$$

and let α_2 be the associated separating curve. Since YZ is of type 1 or 2, Z contains β_1 and β_2 .

If $\alpha_2 \in Z$ and

$$YZ: \alpha_3 \longrightarrow \alpha'_3,$$

then by applying the move

$$\{\beta_1, \beta_2\} \xleftarrow{4} \alpha_1$$

to Z we obtain a vertex

$$W = \{\alpha_1, \alpha_2, \alpha'_3, \dots\}$$

such that $XYZW$ is a quadrangle.

Conversely, suppose $XYZW$ is a quadrangle and $\alpha_2 \notin Z$. We have

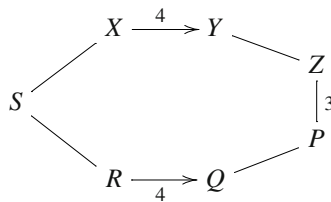
$$X = \{\alpha_1, \alpha_2, \dots\}, \quad Y = \{\beta_1, \beta_2, \alpha_2, \dots\}, \quad Z = \{\beta_1, \beta_2, \alpha'_2, \dots\},$$

for some α'_2 such that $i(\alpha'_2, \alpha_2) = 2$ (YZ is of type 2 since α_2 is separating). Since $W \neq Y$, we have $\alpha_2 \notin W$, and thus $\alpha_1 \in W$ (because $\alpha_1 \in X$). Thus ZW is a move of type 4 of the form

$$ZW: \{\beta_1, \beta_2\} \xleftarrow{4} \alpha_1.$$

and $W = \{\alpha_1, \alpha'_2, \dots\}$. By Remark 5.14, the separating curves associated to ZW and XY are isotopic, so $\alpha_2 \in W \cap Z$, which is a contradiction, as $i(\alpha'_2, \alpha_2) = 2$. $\square$

Definition 5.16 *Standard heptagon* is a circuit of the form



such that the path $X \xrightarrow{4} Y \longrightarrow Z$ is not contained in a quadrangle.

Remark 5.17 By Corollary 4.25, automorphisms of $\mathcal{P}(N)$ map standard heptagons to standard heptagons.

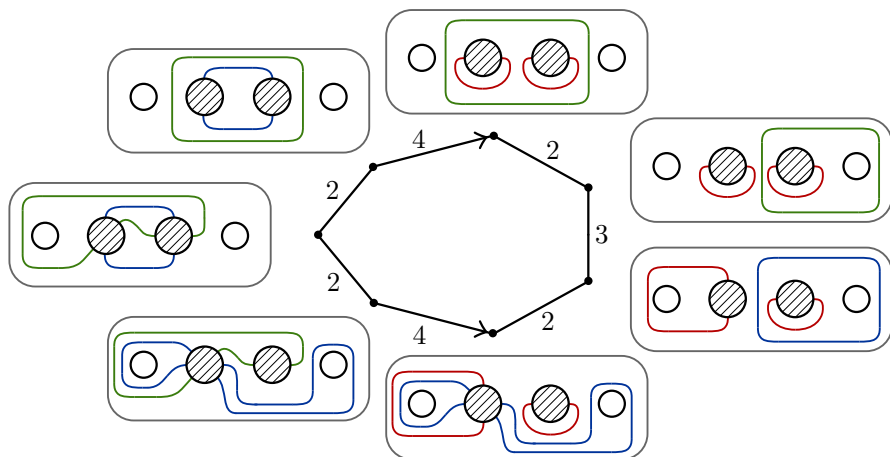
The circuit in Fig. 8 is a standard heptagon. It is easy to check, using Lemma 5.15, that it satisfies the definition.

Lemma 5.18 *Every standard heptagon is 2-tight and alternating.*

Proof Let $\mathcal{C}$ be a standard heptagon as in Definition 5.16 and let

$$XY: \alpha_1 \xrightarrow{4} \{\beta_1, \beta_2\}$$

Since YZ is of type 1 or 2, Y contains at least one two-sided curve, which implies $N \notin \{N_{2,1}, N_3\}$. Let $\alpha_2 \in X \cap Y$ be the separating curve associated to XY . We will show that $X \setminus \{\alpha_1, \alpha_2\}$ is contained in every vertex of $\mathcal{C}$.


 Fig. 8 Standard heptagon in $\mathcal{P}(N_{2,2})$

By definition of standard heptagon and Lemma 5.15, we have $\alpha_2 \notin Z$. Moreover, α_2 is separating, so

$$YZ: \alpha_2 \xrightarrow{2} \alpha_2''$$

for some $\alpha_2'' \in Z$ such that $i(\alpha_2, \alpha_2'') = 2$.

By looking at one-sided curves we see that

$$\begin{aligned} Z^- &= Y^- = X^- \cup \{\beta_1, \beta_2\} = S^- \cup \{\beta_1, \beta_2\}, \\ P^- &= Q^- = R^- \cup \{\beta_1', \beta_2'\} = S^- \cup \{\beta_1', \beta_2'\}. \end{aligned}$$

for some $\beta_1', \beta_2' \in P^-$. But ZP is an edge of type 3, so we may assume that $\beta_1' = \beta_1$ and

$$\begin{aligned} ZP: \beta_2 &\xrightarrow{3} \beta_2', \\ QR: \{\beta_1, \beta_2'\} &\xleftarrow{4} \alpha_1' \end{aligned}$$

for some $\beta_2' \in P^-$ such that $i(\beta_2, \beta_2') = 1$ and $\alpha_1' \in R^+$.

Let $\alpha_2' \in Q \cap R$ be the separating curve associated to QR . By Remark 5.14, α_2' (respectively α_2) is the boundary of a regular neighbourhood of $\alpha_1' \cup \beta_1$ (respectively $\alpha_1 \cup \beta_1$). It follows that

$$\alpha_2' = \alpha_2 \iff \alpha_1' = \alpha_1.$$

Suppose that $\alpha_1' = \alpha_1$. Then $\alpha_2 = \alpha_2' \in R \cap Q$. We have $\alpha_2'' \in P$ and $i(\alpha_2'', \alpha_2) = 2$, so $\alpha_2 \notin P$ and $\alpha_2'' \notin Q$. It follows that

$$PQ: \alpha_2'' \xrightarrow{2} \alpha_2$$

We see that Y and Q differ by the move $\beta_2 \rightarrow \beta_2'$, hence $X = R$, a contradiction (with the definition of a circuit).

- (1) $|X| \leq |X'|$ (i.e. if this is an edge of type 4, then α_2 is two-sided and $\underline{\alpha_2}'$ is a pair of disjoint one-sided curves);
- (2) if M is the union of non-trivial components of $N_{X \setminus \{\alpha_1, \alpha_2\}}$ then $M \neq N_{2,1}$, and if $M \in \{S_{1,2}, N_{2,2}\}$ then at least one of the curves $\alpha_1, \alpha_2, \underline{\alpha_2}'$ is separating in M .

Then the edge XX' is contained in a tame circuit $\mathcal{C}$

$$V - X - X' - W - \dots - V,$$

such that

- (1) the path $V - X - X' - W$ is alternating;
- (2) $\alpha_1 \notin V \cup W$;
- (3) X is minimal in the path $V - X - X' - W$.

Proof Let M be the union of non-trivial components of $N_{X \setminus \{\alpha_1, \alpha_2\}}$.

Case 1: M is disconnected. If M_1 and M_2 are connected components of M , then

$$M_1, M_2 \in \{S_{0,4}, S_{1,1}, N_{1,2}, N_{2,1}\}$$

and XX' is contained in an alternating quadrangle $\mathcal{C}$ of the form

$$\begin{array}{ccc} \{\alpha_1, \alpha_2\} & \text{---} & \{\alpha_1, \underline{\alpha_2}'\} \\ | & & | \\ \{\underline{\alpha_1}', \alpha_2\} & \text{---} & \{\underline{\alpha_1}', \underline{\alpha_2}'\} \end{array}$$

Note that $\underline{\alpha_1}'$ and $\underline{\alpha_2}'$ may be pairs of disjoint one-sided curves (corresponding to edges of types 4), but in each case X is minimal in $\mathcal{C}$.

Case 2: M is connected and $M \setminus \{\alpha_1, \alpha_2\}$ has three connected components. In this case $M \cong S_{0,5}$, and we can use standard 5-cell to construct $\mathcal{C}$ – see Fig. 9.

Since $\mathcal{C}$ has no edges of type 4, X is minimal in $\mathcal{C}$.

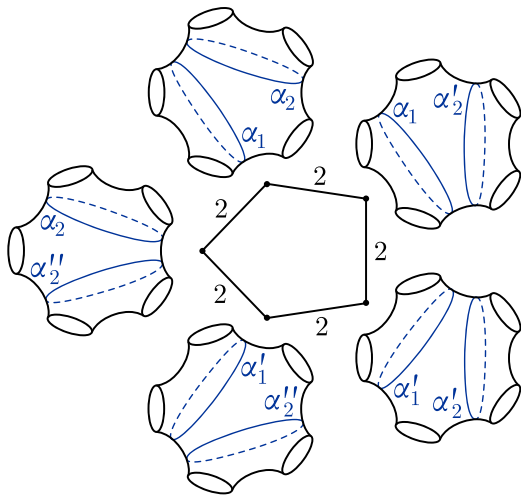
Case 3: M is connected and $M \setminus \{\alpha_1, \alpha_2\}$ has less than 3 connected components.

If $M \setminus \{\alpha_1, \alpha_2\}$ has less than 3 connected components, then it has exactly 2 connected components. Indeed, otherwise $M = N_{2,1}$ or $M = N = N_3$, which contradicts our assumptions.

If $M \cong S_{1,2}$, then the situation is identical to that in the proof of Lemma 6 of [12]. Following the same argument, we obtain a hexagon $\mathcal{C}$ without edges of type 3 or 4 – see Fig. 10. This hexagon is not alternating, and hence $\mathcal{C}$ is tame and X is minimal in $\mathcal{C}$.

Suppose now that $M \cong N_{1,3}$. If α_1 is two-sided, then it is separating in M and $\alpha_2, \underline{\alpha_2}' = \alpha_2'$ are one-sided curves in $M \setminus \{\alpha_1\} \cong N_{1,2}$ connected by an edge of type 3. If α_1 is one-sided, then $\alpha_2, \underline{\alpha_2}' = \alpha_2'$ are two-sided curves in $M \setminus \{\alpha_1\} \cong S_{0,4}$ connected by an edge of type 2. In either case XX' is contained in an alternating hexagon $\mathcal{C}$ without edges of type 4 and with 3 edges of type 3 – see Fig. 11.

Fig. 9 Standard pentagon in $\mathcal{P}(S_{0,5})$



Such $\mathcal{C}$ is tame, and X is minimal in $\mathcal{C}$.

Finally, suppose that $M \cong N_{2,2}$. We assumed that one of α_1 , α_2 or α'_2 is separating. Moreover, $M \setminus \{\alpha_1, \alpha_2\}$ are 2 pairs of pants, so α_1, α_2 or α'_2 cuts $N_{2,1}$ from M .

If α_1 cuts $N_{2,1}$ from M , then $\alpha_2, \alpha'_2 \in N_{2,1}$ are connected by an edge of type 3 or 4. In either case XX' is contained in a standard heptagon $\mathcal{C}$. If XX' is of type 4, then X is minimal in $\mathcal{C}$.

If α_2 (or α'_2) cuts $N_{2,1}$ from M , then α_1 is the unique two-sided curve in $N_{2,1}$ and $\alpha_2, \alpha'_2 = \alpha'_2$ are connected by an edge of type 2. As in the previous case, XX' is contained in a standard heptagon $\mathcal{C}$ and X is minimal in $\mathcal{C}$ (in this case the edges VX and $X'W$ are of different types: one is of type 4 and the other one is of type 2). $\square$

Proposition 6.5 *Let*

$$X = \{\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n\} \text{ --- } \{\alpha_1, \alpha'_2, \alpha_3, \dots, \alpha_n\} = X'$$

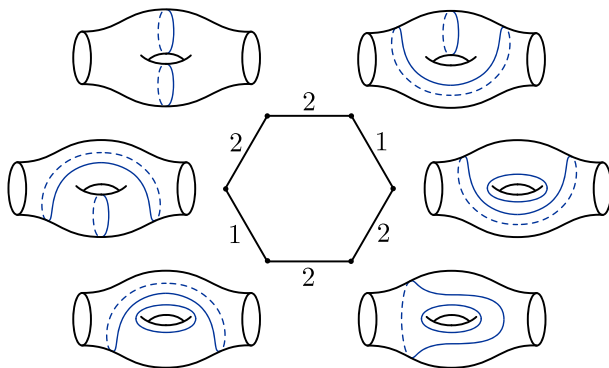


Fig. 10 Standard hexagon in $\mathcal{P}(S_{1,2})$

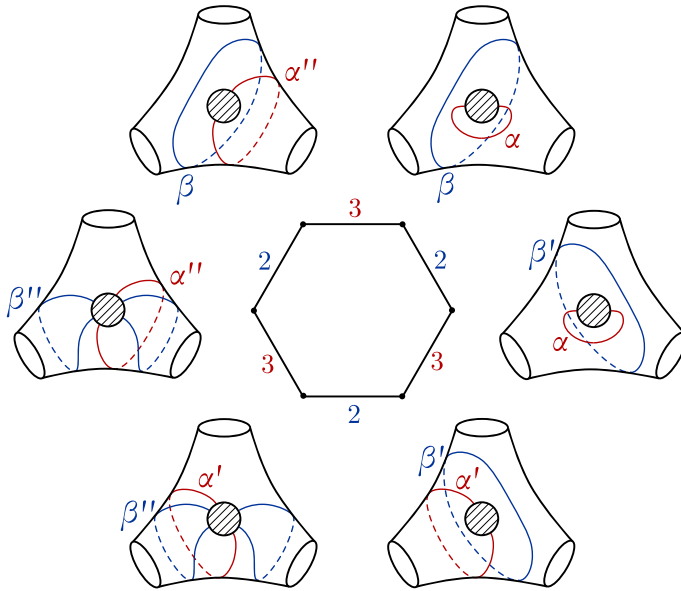


Fig. 11 Tame alternating hexagon in $\mathcal{P}(N_{1,3})$

be an edge in $\mathcal{P}(N)$ and let M be the union of non-trivial components of $N_{X \setminus \{\alpha_1, \alpha_2\}}$. If $M \in \{S_{1,2}, N_{2,2}\}$ and each of $\{\alpha_1, \alpha_2, \underline{\alpha_2'}\}$ is nonseparating in M , then $\underline{\alpha_2'} = \alpha_2'$ is a curve and there exists a vertex

$$X'' = \{\alpha_1, \alpha_2'', \alpha_3, \dots, \alpha_n\}$$

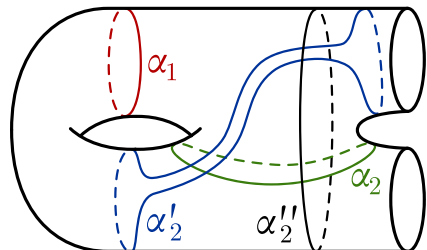
in $\mathcal{P}(N)$ and two edges

$$\begin{aligned} XX'' &: \{\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n\} \text{ --- } \{\alpha_1, \alpha_2'', \alpha_3, \dots, \alpha_n\} \\ X''X' &: \{\alpha_1, \alpha_2'', \alpha_3, \dots, \alpha_n\} \text{ --- } \{\alpha_1, \alpha_2', \alpha_3, \dots, \alpha_n\} \end{aligned}$$

such that XX'' and $X''X'$ satisfy the assumptions of Proposition 6.4.

Proof If $M \cong S_{1,2}$, the construction of α_2'' is given in Case 4 of the proof of Lemma 6 in [12]; see also Fig. 12. We therefore restrict attention to the case $M \cong N_{2,2}$.

Fig. 12 Construction of α_2'' when $M \cong S_{1,2}$



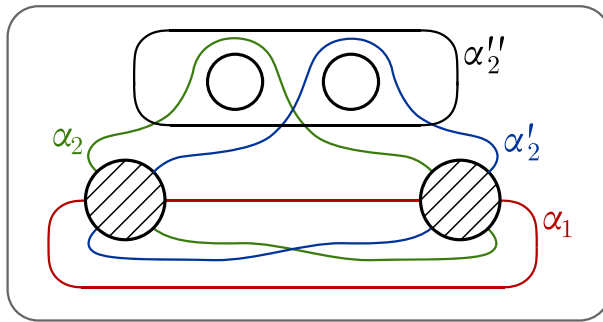


Fig. 13 Construction of α_2'' when $M \cong N_{2,2}$

Note the following:

- (1) $\{\alpha_1, \alpha_2\}$ and $\{\alpha_1, \alpha_2'\}$ decompose M into pairs of pants, hence α_1, α_2 and $\alpha_2' = \alpha_2'$ are two-sided curves;
- (2) α_2 and α_2' intersect in two points in $M \setminus \{\alpha_1\} \cong S_{0,4}$.

Therefore, the configuration of $\alpha_1, \alpha_2, \alpha_2'$ is as shown in Fig. 13. The curve α_2'' is constructed as indicated in the same figure. $\square$

7 Automorphisms of $\mathcal{P}(N)$

Definition 7.1 If (F, X) is a marked F -graph and $\alpha \in X$ is the unique curve such that $F = F(X, \alpha)$ (see Proposition 4.19), then define

$$v[F, X] = \alpha.$$

Remark 7.2 By definition, if $X \in \mathcal{P}^0(N)$ and $\alpha \in X$, then

$$v[F(X, \alpha), X] = \alpha.$$

Definition 7.3 Let $A \in \text{Aut}(\mathcal{P}(N))$ and $\alpha \in \text{Curv}^0(N)$. Extend α to a vertex X of $\mathcal{P}(N)$ and define

$$\Phi(A)(\alpha) = v[A(F(X, \alpha)), A(X)].$$

Proposition 7.4 Definition of Φ does not depend on $X \in \mathcal{P}^0(N)$, hence $\Phi(A)$ is a well defined map

$$\Phi(A) : \text{Curv}^0(N) \rightarrow \text{Curv}^0(N).$$

Proof We need to show that if $X, X' \in \mathcal{P}^0(N)$ are such that $\alpha \in X \cap X'$, then

$$v[A(F(X, \alpha)), A(X)] = v[A(F(X', \alpha)), A(X')].$$

Since $\mathcal{P}(N_\alpha)$ is path-connected by Theorem 3.1, it suffices to show the statement for X and X' being connected by an edge $X - X'$. We can assume that

$$\begin{aligned} X &= \{\alpha, \alpha_2, \alpha_3, \dots, \alpha_n\}, \\ X' &= \{\alpha, \underline{\alpha}_2', \alpha_3, \dots, \alpha_n\}, \end{aligned}$$

so if X and X' are connected by an edge of type 4, we assume that α_2 is two-sided. Let M be the union of non-trivial components of $N_{X \setminus \{\alpha, \alpha_2\}}$. We consider three cases.

Case 1: $N \neq N_3$ and $M \neq N_{2,1}$. By Proposition 6.5, we can assume that X and X' satisfy the assumptions of Proposition 6.4 – if this is not the case, we split the edge

$$X - X'$$

into two edges

$$X - X'' - X'$$

satisfying the assumptions of Proposition 6.4.

By Proposition 6.4, X and X' are consecutive vertices of a tame circuit $\mathcal{C}$ in $\mathcal{P}(N)$, containing an alternating path P

$$V - X - X' - W$$

such that $\alpha \notin V \cup W$, and X is minimal in P . Since, $\alpha \notin V \cup W$ and $\alpha \in X \cap X'$, by Corollary 4.20,

$$\begin{aligned} F(XV) &= F(X, \alpha) \\ F(X'W) &= F(X', \alpha). \end{aligned} \tag{7.1}$$

By Remark 6.2, $A(\mathcal{C})$ is tame and 2-tight. By Remark 5.3, the path $A(P)$

$$A(V) - A(X) - A(X') - A(W)$$

is alternating, and by Lemma 5.8, $A(X)$ is minimal in $A(P)$. By Remark 6.3, $A(P)$ satisfies the assumptions of Lemma 5.9 and we have

$$\begin{array}{ccc} A(V) \supseteq \{\underline{\beta}_1', \beta_2\} & \text{---} & \{\beta_1, \beta_2\} \subseteq A(X) \\ & & \downarrow \\ A(W) \supseteq \{\underline{\beta}_1'', \underline{\beta}_2'\} & \text{---} & \{\beta_1, \underline{\beta}_2'\} \subseteq A(X') \end{array}$$

where β_1 and β_2 are curves. By Corollary 4.20,

$$\begin{aligned} F(A(X)A(V)) &= F(A(X), \beta_1) \\ F(A(X')A(W)) &= F(A(X'), \beta_1). \end{aligned} \tag{7.2}$$

Hence, Remark 7.2, Corollary 4.22, Eqs. 7.1 and 7.2 imply that

$$\begin{aligned} v[A(F(X, \alpha)), A(X)] &= v[A(F(XV)), A(X)] \\ &= v[F(A(X)A(V)), A(X)] = v[F(A(X), \beta_1), A(X)] = \beta_1 \\ v[A(F(X', \alpha)), A(X')] &= v[A(F(X'W, X')), A(X')] \\ &= v[F(A(X')A(W)), A(X')] = v[F(A(X'), \beta_1), A(X')] = \beta_1. \end{aligned}$$

Case 2: $M = N_{2,1}$. In this case α and α_2 are one-sided curves contained in a one-holed Klein bottle M . Let $V, W \in \mathcal{P}^0(N)$ be such that

$$F(X, \alpha) = XV, \quad F(X', \alpha) = X'W.$$

We have a path P in $\mathcal{P}(N)$:

$$V \xrightarrow{3} X \xrightarrow{3} X' \xrightarrow{3} W,$$

which is contained in a fan subgraph. Then $A(P)$ is also a path of edges of type 3 contained in a fan subgraph of $\mathcal{P}(N)$, and by the structure of $\mathcal{P}(N_{2,1})$ (see Proposition 3.4) it is of the same form as in Case 1 with one-sided $\beta_1, \beta_2, \beta'_1, \beta''_1$ and β'_2 . The rest of the proof is the same as in Case 1.

Case 3: $N = N_3$. We can assume that one of the curves α or α_2 is two-sided, because otherwise we are in Case 2. We have $X = \{\alpha, \alpha_2\}$. Extend XX' to an alternating path P

$$V \text{ --- } X \text{ --- } X' \text{ --- } W$$

such that $\alpha \notin V \cup W$. By Remark 5.3, the path $A(P)$ is alternating. Since one of the edges $A(XX')$ or $A(XV)$ is not of type 3 (because α or α_2 is two-sided), $A(X)$ contains a two-sided curve and $|A(X)| = 2$, which means that $A(P)$ is 2-tight and $A(X)$ is minimal. By Lemma 5.9, $A(P)$ is of the same form as in Case 1, and the rest of the proof is the same as in Case 1. $\square$

The key step in the proof Theorem 1.4 is the following result.

Theorem 7.5 *For every nonorientable surface N with $\chi(N) < 0$,*

$$\Phi: \text{Aut}(\mathcal{P}(N)) \rightarrow \text{Aut}(\text{Curv}(N))$$

is an isomorphism.

The proof of the above theorem is divided into several steps, given in Lemmas 7.6 through 7.9.

Lemma 7.6 *If $A \in \text{Aut}(\mathcal{P}(N))$ and $\alpha, \beta \in \text{Curv}^0(N)$ are such that $i(\alpha, \beta) = 0$, then*

$$i(\Phi(A)(\alpha), \Phi(A)(\beta)) = 0.$$

Hence,

$$\Phi(A): \text{Curv}(N) \rightarrow \text{Curv}(N)$$

is a simplicial map.

Proof We extend $\{\alpha, \beta\}$ to a vertex

$$X = \{\alpha, \beta, \gamma_3, \dots, \gamma_n\} \in \mathcal{P}^0(N).$$

Then

$$X = F(X, \alpha) \cap F(X, \beta)$$

and

$$\begin{aligned}\Phi(A)(\alpha) &= v[A(F(X, \alpha)), A(X)] \\ \Phi(A)(\beta) &= v[A(F(X, \beta)), A(X)].\end{aligned}$$

Hence,

$$\Phi(A)(\alpha), \Phi(A)(\beta) \in A(X)$$

are disjoint as they are in the same vertex $A(X)$ of $\mathcal{P}^0(N)$. □

Lemma 7.7 *If $A, B \in \text{Aut}(\mathcal{P}(N))$ and $\alpha \in \text{Curv}^0(N)$, then*

$$\Phi(A)(\Phi(B)(\alpha)) = \Phi(AB)(\alpha).$$

Hence

$$\Phi: \text{Aut}(\mathcal{P}(N)) \rightarrow \text{Aut}(\text{Curv}(N))$$

is a homomorphism.

Proof Let $X \in \mathcal{P}^0(N)$ such that $\alpha \in X$ and $F = F(X, \alpha)$. If

$$\beta = \Phi(B)(\alpha) = v(B(F), B(X)),$$

then $\beta \in B(X)$ and

$$B(F) = F(B(X), \beta).$$

Hence,

$$\begin{aligned}\Phi(A)(\Phi(B)(\alpha)) &= \Phi(A)(\beta) = v[A(F(B(X), \beta)), A(B(X))] \\ &= v[AB(F), AB(X)] = v[AB(F(X, \alpha)), AB(X)] = \Phi(AB)(\alpha).\end{aligned}$$

This and Lemma 7.6 imply that for any $A \in \text{Aut}(\mathcal{P}(N))$, $\Phi(A)$ is an invertible simplicial map, so $\Phi(A) \in \text{Aut}(\text{Curv}(N))$. □

Lemma 7.8 (Injectivity) *If $\Phi(A) = id$ in $\text{Aut}(\text{Curv}(N))$, then $A = id$ in $\text{Aut}(\mathcal{P}(N))$.*

Proof For every $X \in \mathcal{P}^0(N)$ and $\alpha \in X$ we have

$$\alpha = \Phi(A)(\alpha) = v[A(F(X, \alpha)), A(X)] \in A(X),$$

so $X \subseteq A(X)$. But X is a maximal multicurve, so $X = A(X)$. $\square$

Lemma 7.9 (Surjectivity) *For every nonorientable surface N with $\chi(N) < 0$,*

$$\Phi: \text{Aut}(\mathcal{P}(N)) \rightarrow \text{Aut}(\text{Curv}(N))$$

is surjective.

Proof Consider the diagram

$$\begin{array}{ccc} \text{Mod}(N) & \xlongequal{\quad} & \text{Mod}(N) \\ \theta \downarrow & & \downarrow \eta \\ \text{Aut}(\mathcal{P}(N)) & \xrightarrow{\quad \Phi \quad} & \text{Aut}(\text{Curv}(N)) \end{array} \quad (7.3)$$

By Theorem 1.3, η is surjective, so it is enough to prove that this diagram commutes. Let $f \in \text{Mod}(N)$, $\alpha \in \text{Curv}^0(N)$ and extend α to

$$X = \{\alpha, \alpha_2, \dots, \alpha_n\} \in \mathcal{P}^0(N).$$

Then

$$\theta(f)(X) = \{f(\alpha), f(\alpha_2), \dots, f(\alpha_n)\} = f(X)$$

and

$$\theta(f)(F(X, \alpha)) = F(f(X), f(\alpha)).$$

By Remark 7.2,

$$\Phi(\theta(f))(\alpha) = v[F(f(X), f(\alpha)), f(X)] = f(\alpha) = \eta(f)(\alpha).$$

$\square$

Proof of Theorem 1.4 Theorem 1.4 follows from commutativity of the diagram (7.3), Theorem 1.3 and the fact that $\Phi: \text{Aut}(\mathcal{P}(N)) \rightarrow \text{Aut}(\text{Curv}(N))$ is an isomorphism – see Theorem 7.5. $\square$

Remark 7.10 Theorem 1.4 has been proved assuming Theorem 1.3. As explained in the introduction, Theorem 1.3 is already known except for the surfaces $N_{2,2}$, $N_{3,1}$, and $N_{4,0}$. The proof for these remaining cases is given in Section 9. This proof is independent of all other results proved in the present paper; in particular, it does not rely on Theorem 1.4.

8 Isomorphisms of Pants Graphs

In this section we prove that if two nonorientable surfaces have isomorphic pants graphs then they are homeomorphic (Theorem 8.4).

For $A \in \mathcal{P}^0(N)$ the number $|A^-|$ is called *one-sided degree* of A and denoted by $d^-(A)$.

Remark 8.1 The maximum one-sided degree of a vertex of $\mathcal{P}(N)$ is equal to the genus of N .

Lemma 8.2 *Let $A \in \mathcal{P}^0(N)$. There are exactly $d^-(A)$ vertices in $\mathcal{P}(N)$ connected to A by an edge of type 3.*

Proof For every $\alpha \in A^-$ there is a unique one-sided curve α' , disjoint from $A \setminus \{\alpha\}$ and such that $i(\alpha, \alpha') = 1$. Indeed, the nontrivial component of the complement of $A \setminus \{\alpha\}$ in N is a two-holed projective plane, and there are only two isotopy classes of nontrivial curves in $N_{1,2}$. $\square$

Lemma 8.3 *If A is vertex of $\mathcal{P}(N_{g,b})$ such that $|A^-| = g$ then A belongs to exactly $g + b - 3$ Farey subgraphs.*

Proof The complement of A^- in $N_{g,b}$ is a $(g + b)$ -holed sphere, so A^+ consists of $g + b - 3$ non-special two-sided curves. If A belongs to a Farey subgraph F then $F = F(A, \alpha)$ for a unique $\alpha \in A^+$ (Proposition 4.19). It follows that A belongs to exactly $g + b - 3$ Farey subgraphs. $\square$

Theorem 8.4 *If N and N' are nonorientable surfaces such that $\mathcal{P}(N)$ is isomorphic to $\mathcal{P}(N')$ then N and N' are homeomorphic. Furthermore, every isomorphism of pants graphs is induced by a homeomorphism of the underlying surfaces.*

Proof Let $N = N_{g,b}$, $N' = N_{g',b'}$ and let $\varphi: \mathcal{P}(N) \rightarrow \mathcal{P}(N')$ be an isomorphism. Since φ preserves type of edges (Corollary 4.25), it also preserves one-sided degree of vertices by Lemma 8.2. Let A be a vertex of $\mathcal{P}(N)$ of maximum one-sided degree. Then $\varphi(A)$ is also a vertex of maximum one-sided degree in $\mathcal{P}(N')$ and

$$g = d^-(A) = d^-(\varphi(A)) = g'.$$

By Lemma 8.3, A belongs to $g + b - 3$ Farey subgraphs, whereas $\varphi(A)$ belongs to $g' + b' - 3$ Farey subgraphs. Since φ maps Farey subgraphs to Farey subgraphs, we have

$$g + b - 3 = g' + b' - 3,$$

which implies $b = b'$.

The second part of the statement follows from Theorem 1.4. $\square$

9 Automorphisms of $\text{Curv}(N)$

The purpose of this section is to complete the description of $\text{Aut}(\text{Curv}(N))$ given in [2] by proving the following result.

Theorem 9.1 *Let $N = N_{g,4-g}$, where $1 \leq g \leq 4$. The natural map*

$$\text{Mod}(N) \rightarrow \text{Aut}(\text{Curv}(N))$$

is an isomorphism.

The *star* of a vertex $\alpha \in \text{Curv}^0(N)$, denoted by $\text{St}(\alpha)$ is the subcomplex of $\text{Curv}(N)$ consisting of all simplices containing α and all faces of such simplices. The *link* of α , denoted by $\text{Lk}(\alpha)$ is the subcomplex of $\text{St}(\alpha)$ consisting of all simplices of $\text{St}(\alpha)$ not containing α . We have $\text{Lk}(\alpha) \cong \text{Curv}(N_\alpha)$. Following [2], we say that a one-sided curve α is *essential* if either $g = 1$ or N_α is nonorientable. We denote by $\text{Curv}_e^0(N)$ the set of essential one-sided vertices of $\text{Curv}(N)$, and for $g \geq 2$ we denote by $\text{Curv}_e^1(N)$ the subgraph of $\text{Curv}^1(N)$ spanned by $\text{Curv}_e^0(N)$.

Lemma 9.2 *For $g \geq 2$ and $b \geq 0$ the graph $\text{Curv}_e^1(N)$ is connected.*

Proof We use the graph $X(N)$ defined in [2], which has $\text{Curv}_e^0(N)$ as the vertex set and two vertices α, β are joined by an edge in $X(N)$ if and only if $i(\alpha, \beta) = 1$. If $\alpha, \beta \in \text{Curv}_e^0(N)$ are such that $i(\alpha, \beta) = 1$ then the complement of a regular neighbourhood of $\alpha \cup \beta$ in N is nonorientable, and so there is an essential one-sided curve γ disjoint from α and β . It follows that if two vertices are connected by a path in $X(N)$ then they are also connected by a path in $\text{Curv}_e^1(N)$. From this observation and the description of connected components of $X(N)$ given in [2] it follows immediately that $\text{Curv}_e^1(N)$ is connected. $\square$

Proposition 9.3 (Theorem 5.1 of [2]) *If $\alpha \in \text{Curv}_e^0(N)$ and $A \in \text{Aut}(\text{Curv}(N))$ then α and $A(\alpha)$ are in the same $\text{Mod}(N)$ -orbit.*

We are ready to prove the main result of this section.

Proof of Theorem 9.1 We use induction on the genus of N . For $g = 1$ the theorem is true by [15]. Let $g \geq 2$ and assume the theorem is true for surfaces of lower genus. Let $A \in \text{Aut}(\text{Curv}(N))$. We need to show that there is a unique $f \in \text{Mod}(N)$ such that $f(\alpha) = A(\alpha)$ for every $\alpha \in \text{Curv}^0(N)$. We break the proof into three steps.

Step 1. *For every $\alpha \in \text{Curv}_e^0(N)$ there is a unique $f_\alpha \in \text{Mod}(N)$ such that $f_\alpha(\beta) = A(\beta)$ for every $\beta \in \text{St}(\alpha)$.*

Let $\alpha \in \text{Curv}_e^0(N)$. By Proposition 9.3, we have $A(\alpha) \in \text{Curv}_e^0(N)$

$$N_{A(\alpha)} \cong N_\alpha \cong N_{g-1,5-g}.$$

By induction hypothesis there is a unique $f'_\alpha: N_\alpha \rightarrow N_{A(\alpha)}$ such that $f'_\alpha(\beta) = A(\beta)$ for every $\beta \in \text{Lk}(\alpha) = \text{Curv}(N_\alpha)$. We need to show that f'_α can be extended over N to a unique $f_\alpha \in \text{Mod}(N)$ such that $f_\alpha(\alpha) = A(\alpha)$. It suffices to show that f'_α maps the boundary component of N_α corresponding to α to the boundary component of $N_{A(\alpha)}$ corresponding to $A(\alpha)$, which is equivalent to $f'_\alpha(\partial N) = \partial N$. There is nothing to prove if $g = 4$, since then $\partial N = \emptyset$.

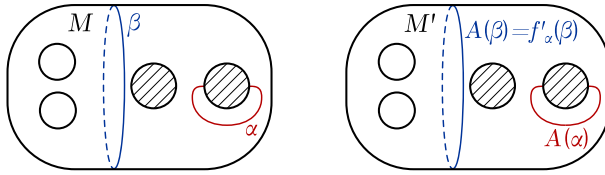


Fig. 14 Curves in $N_{2,2}$ from the proof of Step 1

Assume $g = 2$ and let β be a separating curve disjoint from α such that one of the components of N_β , say M , is a pair of pants containing ∂N (Fig. 14). By Proposition 9.3, $A(\beta)$ is also a separating curve disjoint from $A(\alpha)$ such that one of the components of $N_{A(\beta)}$, say M' , is a pair of pants containing ∂N . Observe that M (respectively M') is the unique orientable component of $N_{\{\alpha, \beta\}}$ (respectively $N_{\{A(\alpha), A(\beta)\}}$). Since $A(\beta) = f'_\alpha(\beta)$ we have $f'_\alpha(M) = M'$ and $f'_\alpha(\partial N) = \partial N$.

Finally, assume $g = 3$ and let β be a separating curve disjoint from α such that both components of N_β are nonorientable and α is contained in the genus 2 component of N_β . Let M be the genus 1 component of N_β , γ a one-sided curve in M , and δ the unique two-sided nonseparating curve in $N \setminus M$ (Fig. 15).

By Proposition 9.3, $A(\beta)$ is also a separating curve disjoint from $A(\alpha)$ such that both components of $N_{A(\beta)}$ are nonorientable. Let M' be the genus 1 component of $N_{A(\beta)}$ which must contain ∂N . By Proposition 9.3, $A(\delta)$ is a two-sided nonseparating curve disjoint from $A(\beta)$, so it must be contained in $N \setminus M'$. Furthermore, $A(\delta)$ fills $N \setminus M'$ which implies that $A(\gamma)$ is contained in M' . Since $A(\beta) = f'_\alpha(\beta)$ and $A(\gamma) = f'_\alpha(\gamma)$ we have $f'_\alpha(M) = M'$ and $f'_\alpha(\partial N) = \partial N$.

Step 2. For every $\alpha, \beta \in \text{Curv}_e^0(N)$ we have $f_\alpha = f_\beta$.

Since $\text{Curv}_e^1(N)$ is connected by Lemma 9.2 we can assume that α and β are disjoint. Then

$$f_\alpha(\alpha) = A(\alpha) = f_\beta(\alpha), \quad f_\alpha(\beta) = A(\beta) = f_\beta(\beta).$$

Let $h = f_\alpha \circ f_\beta^{-1}$, $M = N_{\{\alpha, \beta\}}$ and $h' = h|_M$. Observe that h' fixes every vertex of $\text{Curv}(M) = \text{Lk}(\{\alpha, \beta\})$ because $f_\alpha(\gamma) = f_\beta(\gamma) = A(\gamma)$ for every $\gamma \in \text{Lk}(\{\alpha, \beta\})$. If M is nonorientable then $M \cong N_{g-2, 6-g}$ and $h' = id_M$ by the induction hypothesis. If M is orientable then either $g = 2$ and $M \cong S_{0,4}$, or $g = 4$ and $M \cong S_{1,2}$. Let H be the kernel of the map $\text{Mod}(M) \rightarrow \text{Aut}(\text{Curv}(M))$, which is generated by hyperelliptic involutions, $H \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2$ if $M \cong S_{0,4}$, or $H \cong \mathbb{Z}_2$ if $M \cong S_{1,2}$, see [10]. We know that H acts faithfully on the components of ∂M , and in the case of $S_{0,4}$ it acts by even permutations. Since h' fixes the two components of ∂M corresponding to α and β , we have $h' = id_M$. It follows that $h = id$.

After Step 2 we know that there is a unique $f \in \text{Mod}(N)$ such that for every $\beta \in \text{Curv}_e^0(N)$ and $\alpha \in \text{St}(\beta)$ we have $f(\alpha) = A(\alpha)$.

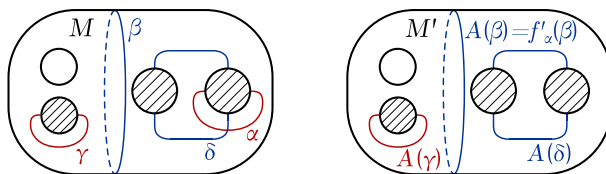


Fig. 15 Curves in $N_{3,1}$ from the proof of Step 1

Step 3. For every $\alpha \in \text{Curv}^0(N)$ we have $f(\alpha) = A(\alpha)$.

We know that $f(\alpha) = A(\alpha)$ if either

- (1) N_α is connected and nonorientable, or
- (2) N_α has two nonorientable components, or
- (3) N_α has a nonorientable component of genus at least 2.

Indeed, in each of these cases there is $\beta \in \text{Curv}_e^0(N)$ such that $\alpha \in \text{St}(\beta)$. It remains to consider the following cases.

- (4) N_α is connected and orientable.
- (5) $N_\alpha \cong S_{1,1} \sqcup N_{1,2}$.

Suppose N_α is connected and orientable. Let β be any curve in N such that $i(\alpha, \beta) = 1$. Observe that β is one-sided and essential, so for every $\gamma \in \text{St}(\beta)$ we have $A(\gamma) = f(\gamma)$. By [2, Lemma 7.5] we also have $A(\gamma) = f(\gamma)$ for every $\gamma \in \text{Curv}^0(N)$ such that $i(\beta, \gamma) = 1$, so in particular $A(\alpha) = f(\alpha)$.

Finally, suppose $N_\alpha \cong S_{1,1} \sqcup N_{1,2}$. Consider the curves β, γ and δ in Fig. 16.

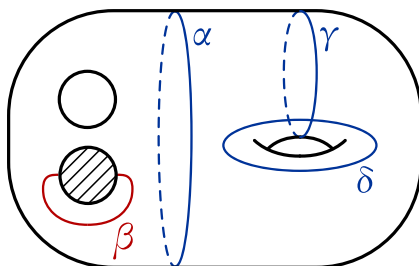
Observe that $\text{Lk}(\beta) \cap \text{Lk}(\gamma) \cap \text{Lk}(\delta) = \{\alpha\}$. We have $f(\beta) = A(\beta)$ by case (4) and $f(\gamma) = A(\gamma)$, $f(\delta) = A(\delta)$ by case (1), which implies

$$\begin{aligned} \{f(\alpha)\} &= \text{Lk}(f(\beta)) \cap \text{Lk}(f(\gamma)) \cap \text{Lk}(f(\delta)) \\ &= \text{Lk}(A(\beta)) \cap \text{Lk}(A(\gamma)) \cap \text{Lk}(A(\delta)) = \{A(\alpha)\}. \end{aligned}$$

□

Remark 9.4 Our proof of Theorem 9.1 is similar to the proof of the main result of [2]. In fact Step 1 of our proof is the same as in [2, Theorem 7.7] (although more detailed in the case $n = 1$). The main difference is in Step 2, where we use connectivity of

Fig. 16 Curves in $N_{3,1}$ from case 5



the graph $\text{Curv}_e^1(N)$, which requires the assumption $g \geq 2$. We can assume $g \geq 2$ because the case of $N_{1,3}$ was settled in [15] by a different argument. The proof in [2] is based on connectivity of a different graph and it works also for $g = 1$.

10 The Ggraph $\mathcal{P}(N_3)$.

In this section we focus on the case $N = N_3$. We describe the structure of the pants graph $\mathcal{P}(N)$ and construct an abstract graph isomorphic to it. Additionally, we define a simply-connected 2-dimensional complex $\overline{\mathcal{P}}(N)$, whose 1-skeleton is $\mathcal{P}(N)$, and such that $\text{Aut}(\overline{\mathcal{P}}(N)) = \text{Aut}(\mathcal{P}(N))$.

10.1 Curves on N

There is a unique (up to isotopy) one-sided curve α_0 in N such that $N_{\alpha_0} \cong S_{1,1}$, see [14]. The remaining curves in N can be partitioned into two disjoint subsets:

$$\text{Curv}^0(N) \setminus \{\alpha_0\} = \text{Curv}^+ \cup \text{Curv}^-,$$

where Curv^+ consists of two-sided curves, and Curv^- consisting of one-sided curves whose complement is nonorientable.

For every $\alpha \in \text{Curv}^0(N)$, $\alpha \neq \alpha_0$

$$N_\alpha \cong \begin{cases} N_{1,2} & \alpha \in \text{Curv}^+ \\ N_{2,1} & \alpha \in \text{Curv}^- \end{cases} \quad i(\alpha, \alpha_0) = \begin{cases} 0 & \alpha \in \text{Curv}^+ \\ 1 & \alpha \in \text{Curv}^- \end{cases}.$$

Lemma 10.1 *There exists a bijection $\varphi: \text{Curv}^- \rightarrow \text{Curv}^+$ such that for $\alpha \in \text{Curv}^-$ and $\beta \in \text{Curv}^+$*

$$i(\alpha, \beta) = 0 \iff \beta = \varphi(\alpha).$$

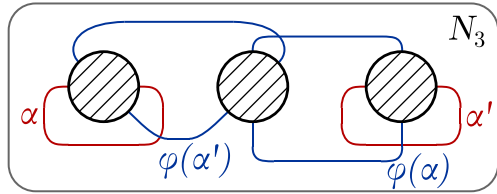
Proof For $\alpha \in \text{Curv}^-$, since $N_\alpha \cong N_{2,1}$, there exists a unique two-sided curve β such that $i(\alpha, \beta) = 0$. We define $\varphi(\alpha) = \beta$. To prove that φ is a bijection, we construct its inverse $\varphi^{-1}: \text{Curv}^+ \rightarrow \text{Curv}^-$. For $\beta \in \text{Curv}^+$, since $N_\beta \cong N_{1,2}$, there exist two one-sided curves disjoint from β . One of them is α_0 , and the other one is $\varphi^{-1}(\beta)$. $\square$

Lemma 10.2 *For $\alpha, \alpha' \in \text{Curv}^-$*

$$i(\alpha, \alpha') = 0 \iff i(\varphi(\alpha), \varphi(\alpha')) = 1.$$

Proof See the configuration of pairs $\{\alpha, \varphi(\alpha)\}$ and $\{\alpha', \varphi(\alpha')\}$ in Fig. 17. $\square$

Fig. 17 Curves $\alpha, \alpha', \varphi(\alpha)$ and $\varphi(\alpha')$ in N_3 – Lemma 10.2



10.2 Vertices and Edges of $\mathcal{P}(N)$.

Proposition 10.3 Every vertex of $\mathcal{P}(N)$ belongs to one of the following sets.

$$V_1 = \{\{\alpha_0, \beta\} \mid \beta \in \text{Curv}^+\},$$

$$V_2 = \{\{\alpha, \varphi(\alpha)\} \mid \alpha \in \text{Curv}^-\} = \{\{\varphi^{-1}(\beta), \beta\} \mid \beta \in \text{Curv}^+\},$$

$$V_3 = \{\{\alpha_1, \alpha_2, \alpha_3\} \mid \alpha_i \in \text{Curv}^-, i = 1, 2, 3, i(\alpha_i, \alpha_j) = 0 \text{ for } i \neq j\}.$$

Proof Let $X \in \mathcal{P}^0(N)$. If $X^+ = \emptyset$ then $X \in V_3$. If $\beta \in X^+$ then since $N_\beta \cong N_{1,2}$, either $X = \{\beta, \alpha_0\} \in V_1$ or $X = \{\beta, \varphi^{-1}(\beta)\} \in V_2$. $\square$

Let $\psi: V_1 \rightarrow V_2$ be the bijection defined by

$$\psi(\{\alpha_0, \beta\}) = \{\varphi^{-1}(\beta), \beta\}$$

for $\beta \in \text{Curv}^+$. For $i = 1, 2$ we define maps $v_i: \text{Curv}^+ \cup \text{Curv}^- \rightarrow V_i$ by the formulas

$$v_1(\alpha) = \begin{cases} \{\alpha_0, \alpha\} & \alpha \in \text{Curv}^+ \\ \{\alpha_0, \varphi(\alpha)\} & \alpha \in \text{Curv}^- \end{cases} \quad v_2(\alpha) = \begin{cases} \{\varphi^{-1}(\alpha), \alpha\} & \alpha \in \text{Curv}^+ \\ \{\alpha, \varphi(\alpha)\} & \alpha \in \text{Curv}^- \end{cases}.$$

Observe that $v_2 = \psi \circ v_1$. For $i = 1, 2$ the restrictions $v_i|_{\text{Curv}^+}$ and $v_i|_{\text{Curv}^-}$ are bijections onto V_i , and for $\alpha \in \text{Curv}^-, \beta \in \text{Curv}^+$ we have

$$v_i(\alpha) = v_i(\beta) \iff \beta = \varphi(\alpha).$$

Proposition 10.4 Every edge of $\mathcal{P}(N)$ belongs to one of the following sets.

$$E_{11} = \{v_1(\beta) \overset{1}{-} v_1(\beta') \mid \beta, \beta' \in \text{Curv}^+, i(\beta, \beta') = 1\},$$

$$E_{12} = \{X \overset{3}{-} \psi(X) \mid X \in V_1\},$$

$$E_{23} = \{X \overset{4}{-} v_2(\alpha) \mid X \in V_3, \alpha \in X\},$$

$$E_{33} = \{\{\alpha_1, \alpha_2, \alpha_3\} \overset{3}{-} \{\alpha'_1, \alpha_2, \alpha_3\} \mid \{\alpha_1, \alpha_2, \alpha_3\}, \{\alpha'_1, \alpha_2, \alpha_3\} \in V_3\}.$$

Proof Let $e = XY$ be an edge of $\mathcal{P}(N)$. If $X, Y \in V_3$ then $e \in E_{33}$. Suppose that e is of the form

$$\{\alpha, \beta\} - \{\alpha', \beta\},$$

10.3 Subgraphs of $\mathcal{P}(N)$.

Consider the following natural subgraphs of $\mathcal{P}(N)$:

$$\Gamma_1 = (V_1, E_{11}), \quad \Gamma_3 = (V_3, E_{33}).$$

Proposition 10.8 Γ_1 is the unique Farey subgraph of $\mathcal{P}(N)$.

Proof We have

$$\Gamma_1 \cong \mathcal{P}(N_{\alpha_0}) \cong \mathcal{P}(S_{1,1}),$$

so Γ_1 is a Farey subgraph. Since Γ_1 contains all edges of type 1, and there are no edges of type 2 in $\mathcal{P}(N)$, Γ_1 contains all Farey triangles of $\mathcal{P}(N)$. $\square$

Definition 10.9 (Farey tree) Farey tree Γ_1^* is the graph which has a vertex for each triangle of Γ_1 , and two vertices of Γ_1^* are joined by an edge if and only if the corresponding triangles share an edge.

As the name suggests, Γ_1^* is a regular tree of degree 3. Let $\mathcal{T}(\Gamma_1)$ denote the set of triangles of Γ_1 . In this section we identify a triangle with its set of vertices. By Lemma 10.2, for $\alpha_i \in \text{Curv}^-$, $i = 1, 2, 3$

$$\{\alpha_1, \alpha_2, \alpha_3\} \in V_3 \iff \{v_1(\alpha_1), v_1(\alpha_2), v_1(\alpha_3)\} \in \mathcal{T}(\Gamma_1).$$

Let us define $\theta: V_3 \rightarrow \mathcal{T}(\Gamma_1)$ by

$$\theta(\{\alpha_1, \alpha_2, \alpha_3\}) = \{v_1(\alpha_1), v_1(\alpha_2), v_1(\alpha_3)\}.$$

Since $v_1|_{\text{Curv}^-}$ is a bijection, θ is also a bijection. Furthermore, for $X, Y \in V_3$, $XY \in E_{33}$ if and only if the triangles $\theta(X)$ and $\theta(Y)$ share an edge. Thus θ is a graph isomorphism $\theta: \Gamma_3 \rightarrow \Gamma_1^*$ and we have the following result.

Proposition 10.10 Γ_3 is isomorphic to the Farey tree Γ_1^* .

Lemma 10.11 For $X \in V_2$ and $Y \in V_3$,

$$XY \in E_{23} \iff \psi^{-1}(X) \in \theta(Y).$$

Proof $XY \in E_{23}$ if and only if $X = v_2(\alpha)$ for some $\alpha \in Y$, in other words $X \in v_2(Y)$. We have $v_2(Y) = \psi(v_1(Y)) = \psi(\theta(Y))$, so $X \in v_2(Y) \iff \psi^{-1}(X) \in \theta(Y)$. $\square$

Proposition 10.12 Every $X \in V_2$ is the centre of a unique fan subgraph of $\mathcal{P}(N)$ with vertex set

$$\{X\} \cup \{Y \in V_3 \mid \psi^{-1}(X) \in \theta(Y)\}.$$

Proof If $X = \{\varphi^{-1}(\beta), \beta\} \in V_2$, where $\beta \in \text{Curv}^+$, then $F = F(X, \beta)$ is the unique fan subgraph of $\mathcal{P}(N)$ containing X . If $Y \neq X$, then Y is a vertex of F if and only if $XY \in E_4$, which is equivalent to $\psi^{-1}(X) \in \theta(Y)$ by the previous lemma. $\square$

10.4 Abstract Model of $\mathcal{P}(N)$

Using the results of the two previous subsections we can construct an abstract graph isomorphic to $\mathcal{P}(N)$ starting from the Farey graph $\mathcal{F}$ and the associated Farey tree $\mathcal{F}^*$. Let $V(\mathcal{F})$ (resp. $V(\mathcal{F}^*)$) denote the vertex set of $\mathcal{F}$ (resp. $\mathcal{F}^*$). Recall that $V(\mathcal{F}^*)$ is the set of triangles of $\mathcal{F}$.

We define $\mathcal{G}$ to be the graph obtained from the disjoint union $\mathcal{F} \sqcup \mathcal{F}^*$ by adding a new vertex $\tilde{x}$ for every $x \in V(\mathcal{F})$ together with edges $\tilde{x}x$ and $\tilde{x}T$ for every $T \in V(\mathcal{F}^*)$ such that $x \in T$. It follows from the results of the two previous subsections that $\mathcal{G}$ is isomorphic to $\mathcal{P}(N)$, with $\mathcal{F}$ and $\mathcal{F}^*$ corresponding to Γ_1 and Γ_3 respectively, and for $x \in V(\mathcal{F})$ corresponding to $X \in V_1$, $\tilde{x}$ corresponds to $\psi(X) \in V_2$.

10.5 Attaching 2-Cells to $\mathcal{P}(N)$.

The analysis from the previous section indicates that there are natural cycles of length 5 (pentagons) in $\mathcal{P}(N)$.

Definition 10.13 For $X \in V_3$ and $\alpha_1, \alpha_2 \in X$ we define a *standard pentagon* to be the circuit

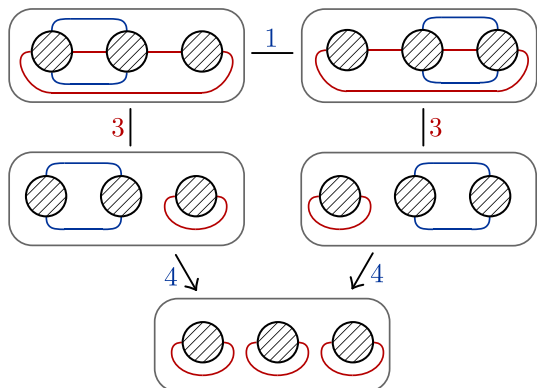
$$X \xleftarrow{4} v_2(\alpha_1) \xrightarrow{3} v_1(\alpha_1) \xrightarrow{1} v_1(\alpha_2) \xrightarrow{3} v_2(\alpha_2) \xrightarrow{4} X.$$

Example 10.14 See Fig. 19 for the geometric realization of a standard pentagon in N .

Theorem 10.15 The 2-dimensional complex $\overline{\mathcal{P}(N)}$ obtained by attaching 2-cells to $\mathcal{P}(N)$ along each triangle and each standard pentagon is simply-connected.

Proof Let L be a loop in $\overline{\mathcal{P}(N)}$ based at $X_0 \in V_1$. We can assume that L is a simplicial loop in the graph $\mathcal{P}(N)$. Since every edge from E_{33} is contained in a fan triangle, every loop is homotopic in $\overline{\mathcal{P}(N)}$ to a loop without edges from E_{32} , so we assume that L does not contain edges from E_{33} . Let n be the length of L , i.e. the number of edges contained in L . We prove by induction on n that L is null-homotopic in $\overline{\mathcal{P}(N)}$. If $n = 0$ then L is obviously null-homotopic, so assume $n \geq 1$ and that all loops of length less than n are null-homotopic.

Fig. 19 Standard pentagon in N_3



If L is contained in the Farey subgraph Γ_1 , then L is null-homotopic, because the complex obtained by attaching 2-cells to Γ_1 along each Farey triangle is simply-connected.

If L is not contained in Γ_1 , then it contains at least one vertex $X_1 \in V_2$ and we can assume that X_1 is joined to X_0 by an edge from E_{12} and this edge is contained in L . If $n = 2$ then L travels from X_0 to X_1 and then back to X_0 along the same edge, so it is null-homotopic. If $n \geq 3$ then L must contain $X_2 \in V_3$, $X_3 \in V_2$, and edges $X_1X_2 \in E_{23}$, $X_2X_3 \in E_{23}$ (because we assumed that L contains no edges from E_{33}). Consider the following segment L' of L :

$$X_0 \xrightarrow{3} X_1 \xrightarrow{4} X_2 \xleftarrow{4} X_3.$$

If $X_3 = X_1$ then by replacing L' by the single edge X_0X_1 we obtain a loop of length $n - 2$ homotopic to L , which is null-homotopic by induction hypothesis. If $X_3 \neq X_1$ then L' is contained in a standard pentagon

$$X_0 \xrightarrow{3} X_1 \xrightarrow{4} X_2 \xleftarrow{4} X_3 \xrightarrow{3} Y \xrightarrow{1} X_0,$$

where $Y = \psi^{-1}(X_3)$. By replacing L' by the segment

$$X_0 \xrightarrow{1} Y \xrightarrow{3} X_3,$$

we obtain a loop of length $n - 1$ homotopic to L , which is null-homotopic by induction hypothesis. $\square$

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Declarations

Conflicts of Interest The authors have no conflict of interest to declare that are relevant to this article.

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